https://doi.org/10.1038/s44435-025-00003-0

A review of smart integrated energy systems towards industrial carbon neutrality: Opportunity and challenge

Check for updates

Yihui Mao[1,4], Nan Zhang[1,4], Pingyuan Shi[1,2,4], Wei Zhong[1,3] & Xiaojie Lin[1,3]

The complex operation of industrial integrated energy systems (IES), with their coupled energy, material, and information flows, is being transformed by artificial intelligence (AI). This paper reviews the AI-driven advancements in IES modeling, design, and operation. We explore the potential of large language models-assisted decision-making and the energy–AI–industrial multidisciplinary nexus. The deep integration of AI and energy science promises to reshape industrial energy management and accelerate the net-zero carbon transition.

The clean and low-carbon transition of the energy system is a key issue in addressing global climate change, environmental governance, and water and food crises. To safeguard a green and livable Earth, the Paris Agreement[1] proposed the goal of keeping the global temperature increase well below 2 °C above pre-industrial levels and pursuing efforts to limit the temperature increase to 1.5 °C. However, the World Meteorological Organization confirmed that global temperature in 2024 has exceeded Paris Agreement targets, rising 1.55 °C above pre-industrial levels[2]. To actively respond to the aforementioned crises, over 100 countries have established carbon neutrality targets[3]. Many developing countries have also introduced policies and laws[4–7] to reduce carbon emissions and promote renewable energy. The advancement of carbon neutrality goals will transform global economic and social systems.

Decarbonizing the industrial sector represents a crucial pathway to achieving carbon neutrality. It also serves as a modern production mode for enterprise clusters that foster collaboration along the industrial chain, promote multi-energy complementarity, and enable resource sharing[8]. China, being the world's largest energy consumer and manufacturing hub, has an energy consumption per unit of GDP in its industrial sector that is 40% higher than that of the United States and almost double that of the European Union[9]. China currently has over 2500 national and provincial-level industrial parks[10]. Industrial activities in the parks account for over 50% of the gross domestic product. However, they also contribute to 31% of China's carbon emissions and 69% of the total social energy consumption. The five major high-energy-consuming process industries (iron and steel, petrochemicals, non-ferrous metals, and chemicals) account for nearly 55% of total industrial energy consumption[11]. China's energy production and industrial energy consumption exhibit an inverse relationship, with varying regional development foundations and resource endowments. Thus, creating integrated energy systems (IES) suited to local conditions is strategically important for achieving green, low-carbon transformation of industrial clusters and supplying diverse, clean, efficient energy to overcome the carbon lock-in effect[12] and reach carbon neutrality goals.

IES is becoming a current research hotspot. IESs make energy use cleaner and more efficient by harmoniously combining different energy sources and coordinating generation, grid, load, and storage. Additionally, they integrate physical infrastructure and digital technologies to achieve precise and intelligent control of multi-energy flows. To investigate the current research landscape, we analyzed 3516 articles (2014–2024) from the Web of Science using VOSviewer, as shown in Fig. 1. The keyword co-occurrence map clearly illustrates that the field is organized into three main thematic clusters. The red cluster shows the core concepts of IESs and their operational optimization. The green cluster focuses on renewable and clean energy sources, which are fundamental to carbon neutrality. The blue cluster encompasses key technologies, such as storage and conversion devices, that provide flexibility. The results show that IES has replaced distributed energy systems (DES)[13], multi-energy systems (MES)[14], microgrid (MG)[15], and others to become a more popular research topic. The central position of the 'operation' keyword, connecting all three clusters, highlights that smart operational strategies are the key to integrating elements such as energy storage, renewable energy, and demand response. This structure confirms that the smart operation of IES is the focal point of current research efforts towards a sustainable energy future, setting the stage for the analysis presented in this review.

The ongoing development of IESs is showcasing new characteristics and challenges as it transforms the energy structure. On one hand, the future energy system, focusing heavily on high penetration of renewable energy, will require enhanced flexibility. In 2023, China's renewable energy installed capacity hit around 1384 GW, exceeding that of coal-fired power for the first time. Zhuo et al.[16] estimated that the total energy potential of wind and solar photovoltaic power in China could reach 200.9 PWh per year, which is 13.5 times China's maximum projected electricity demand in 2050. The daily

[1]College of Energy Engineering, Zhejiang University, Hangzhou, China. [2]Polytechnic Institute, Zhejiang University, Hangzhou, China. [3]Shanghai Institute for Advanced Study, Zhejiang University, Shanghai, China. [4]These authors contributed equally: Yihui Mao, Nan Zhang, Pingyuan Shi. e-mail: xiaojie.lin@zju.edu.cn

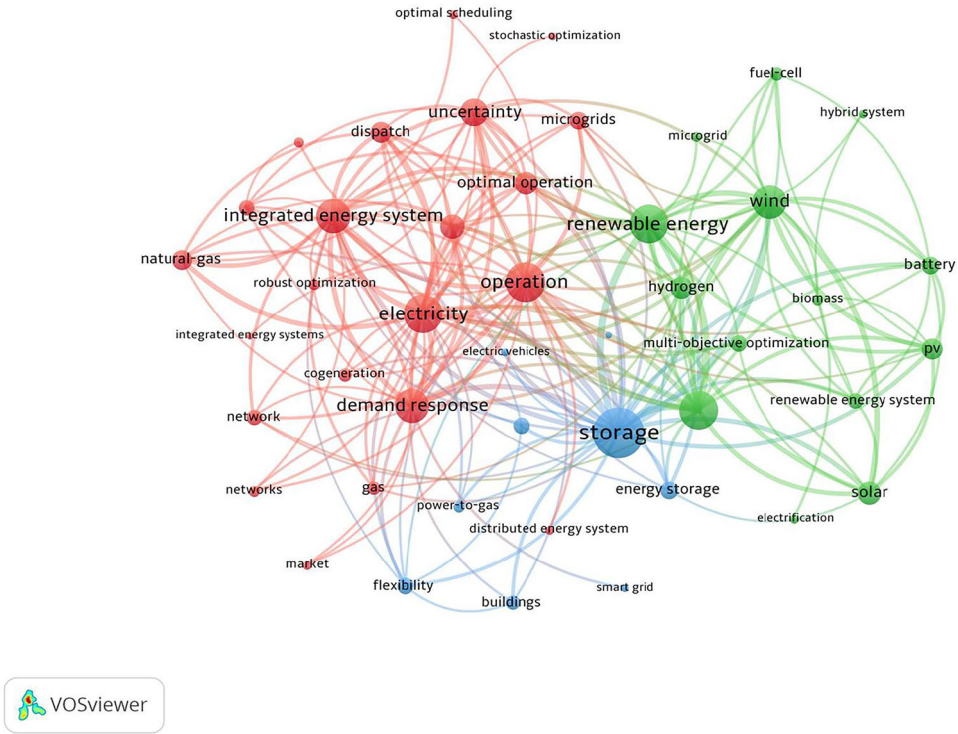

Fig. 1 | Network visualization map of the top 50 keywords returned from a search on integrated energy systems smart operation. The map was created by VOSviewer. 3516 articles were retrieved from the Web of Science from 2014 to 2024.

regulation range of conventional power sources of China's State Grid has exceeded 300 GW. The bi-interaction between the source and load, combined with multi-time scale coupling, poses a considerable challenge for the operation of future IESs. On the other hand, the deep integration of artificial intelligence (AI) technology into IESs is creating new opportunities for energy savings and carbon reduction, in addition to low-carbon green fuels[17–19], energy-efficient conversion devices[20,21], and system energy efficiency improvement[22]. The Global Enabling Sustainability Initiative estimated that digital technologies can reduce global carbon emissions by 20% within the next decade by empowering the energy and industrial sectors[23]. According to Ding et al.[24], the full adoption of AI could potentially reduce building energy use and carbon emissions by ~8–19% by 2050.

However, this enthusiasm masks a critical weakness: current AI cannot inherently guarantee compliance with the fundamental constitutive equations and physical constraints of multi-energy flows. This deficit means AI-driven decisions could be physically unfeasible or even dangerous. Therefore, establishing a regulatory framework for IES that can govern and trust AI-driven control remains a challenge. Future breakthroughs in transformative technologies like AI and large language models (LLMs) are urgently needed to bridge this gap, enabling smart energy systems to achieve fully autonomous and physically reliable multi-energy operations. Figure 2 illustrates the literature on the modeling, design, and operation of IESs integrated with AI technology from 2014 to 2024. AI-related literature has demonstrated a rapid growth trend, from 12 to 286 annual publications. The data also reveals two key trends. First, the field has pivoted from foundational modeling to a dominant focus on practical operation and control, indicating a move toward solving real-world challenges (Fig. 2a). Second, the geographical distribution is led by nations with the most significant energy demands and ambitious carbon neutrality goals, such as China, the US, and India, which accounting for about 90% publications (Fig. 2b).

The integration of AI into energy systems has garnered significant research attention in recent years. Alabi et al.[25] provided a thorough overview of the optimization methods and machine learning (ML) techniques applications in IES, noting that the integrated ML in IES is still in its infancy. Entezari et al.[26] conducted a bibliometric analysis of around 2000 highly cited energy papers. The findings reveal that AI currently impacts only a small portion of energy-related fields but has significant integration potential. Additionally, the number of AI-related patents in the energy sector is 17 times greater than that of scientific papers, highlighting a stronger industrial demand for AI. Moreover, Table 1 provides a list of recent review articles on AI in energy systems, categorized by different system types and focus areas. However, a comprehensive review specifically addressing the modeling, design, and operation of IES in the industrial sector remains scarce. Furthermore, there is a notable lack of discussion on the potential impact of emerging LLMs on IES and its interdisciplinary synergies. Unlike general energy systems, IESs for industrial carbon neutrality face stricter constraints on energy and material flows, stakeholders, environmental factors, and market considerations. Their dynamic nature and uncertainty require consideration of new characteristics, such as energy efficiency, carbon emissions, and economic value. Therefore, this review provides a systematic analysis based on our proposed "Energy + AI" framework, which is structured around five interconnected parts (see Fig. 3): (1) Modeling, which reviews the progression from mechanism-based theories to advanced physics-informed hybrid models; (2) Design, focusing on multi-timescale and decarbonized strategies; (3) Operation, analyzing methods for incorporating energy storage and enhancing multi-sector flexibility; (4) Industrial integration, which specifically addresses the synergistic modeling, design, and operation of coupled energy and production systems; and (5) AI nexus, which explores the frontier from AI embedded and LLM-assisted energy management to the conceptualization and application of the multidisciplinary synergistic.

To bridge the existing gaps, this review provides a unique and critical synthesis of the field. Specifically, the main contributions of this work are threefold:

(1) A systematic review of modeling methods for IESs is presented, covering the evolution from foundational mechanism-based theories to advanced physics-informed hybrid models (e.g., PINN, GNN, and Agent-based). Uniquely, this review also focuses on the integrated modeling of industrial production and energy supply, addressing the challenges of coupling energy and material flows.

(2) With industrial decarbonization as the core objective, typical design and operation strategies for IESs are comprehensively analyzed. We connect

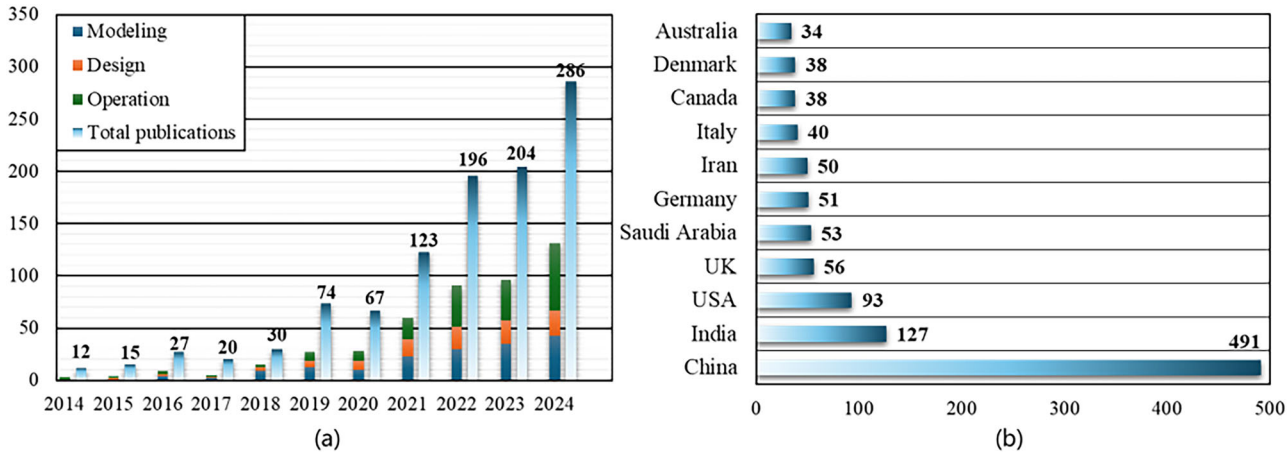

Fig. 2 | Research on AI-related integrated energy systems in the past decade. a Publications on system modeling, planning/design, and operation/control. **b** Publications of articles by country.

Table 1 | Recent review publications about energy system and AI

Ref.	Year	Contents	Aspects
279	2022	Data-driven techniques in integrated energy systems	Load forecast
280	2022	Explainable AI for energy and power systems	Energy management, system monitoring
281	2024	Applications of AI in renewable energy systems	Modeling and optimization
282	2024	AI for renewable power systems operation	Forecasts, dispatch, and control
283	2025	AI for hydrogen-enabled IESs	Energy management, data analytics
284	2025	AI in enhancing urban energy resilience	Resilience evaluation
285	2025	Reinforcement learning for smart grid operation	Uncertainty, safe operation

multi-timescale optimization frameworks with key technologies, such as physical and virtual energy storage and multi-sector flexibility, to address the challenges faced by energy-intensive industries in achieving low-carbon operation.

(3) The frontier of the " Energy and AI" nexus is explored, discussing the transformative potential of emerging technologies, such as AI and LLMs, for optimized decision-making and control. Furthermore, within a multi-disciplinary framework, this review elaborates on the interdisciplinary synergies among AI, energy, and industry and proposes solutions for the future zero-carbon smart factory. Finally, it provides an outlook on the technical bottlenecks and industrial implementation pathways for future IESs, aiming to guide subsequent research and development.

This review systematically organizes the technological landscape and cutting-edge advancements in integrating AI with the modeling, design, and operation of IESs within industrial carbon neutrality. It aims to provide theoretical and technical references for the zero-carbon and intelligent transformation of the industrial sector.

Integrated energy system modeling

Energy system modeling and simulation fundamentally depend on physical device models and the energy flow topology. This is accomplished using computers to simulate the dynamic behaviors and interrelationships among energy device components under various operating conditions, thereby facilitating the optimized design and operation of IESs. In contrast to traditional energy modeling methods, the innovative IES modeling paradigm showcases the integration of diverse components in a region—energy supply, transportation, conversion, storage, and consumption—into a unified system. This addresses the various new demands for clean, low-carbon[27], safety[28], high efficiency[29], and flexibility[30] of IESs under the carbon neutrality goal. This section focuses on mechanism-based modeling, physics-informed hybrid modeling, and energy-production fusion modeling. Figure 4 presents the leading system modeling software tools, open-source frameworks, and theoretical technologies in integrated energy.

Modeling tools and frameworks

A wide array of IES modeling tools and frameworks has been developed to manage the growing complexity of modern energy systems. This complexity arises from the need to integrate renewable energy utilization, demand-side management, and coordinate electricity and carbon markets. As a result, companies and research institutions in the energy, power, petroleum, and chemical sectors have created diverse energy modeling tools. The review by Majidi et al.[31] assessed 97 energy system modeling tools, highlighting a prevalent trend of using the Python programming language for their development, with a significant 89% being open-source and accessible through community collaboration. The evolution of energy system modeling software has undergone significant technological advancements, moving from single-physics field analysis[32,33] to multi-energy flow coupling[34,35], progressing from steady-state simulation to dynamic modeling[36,37], and advancing towards digital twins (DT)[38]. Some tools concentrate on high-precision modeling of specific subsystems. For instance, refs. 39–41 utilized commercial software such as APROS, Aspen Plus, and Ebsilon to create dynamic models of power plant systems and simulate their thermodynamic processes. References [42–44] utilized modeling tools, including EnergyPlus, TRNSYS, and IDA ICE, to execute dynamic simulations, optimize design processes, and undertake life cycle analyses of building energy systems. References [45–47] employed tools such as Dymola, MATPOWER, and CloudPSS to simulate the dynamic transport processes in thermal (including hot water/steam) networks, natural gas networks, and power networks. In recent years, enhanced modeling technologies like AI and DT, along with cloud platforms, have enabled city-level modeling[48,49]. These tools primarily focus on large-scale 3D visualization and energy systems management.

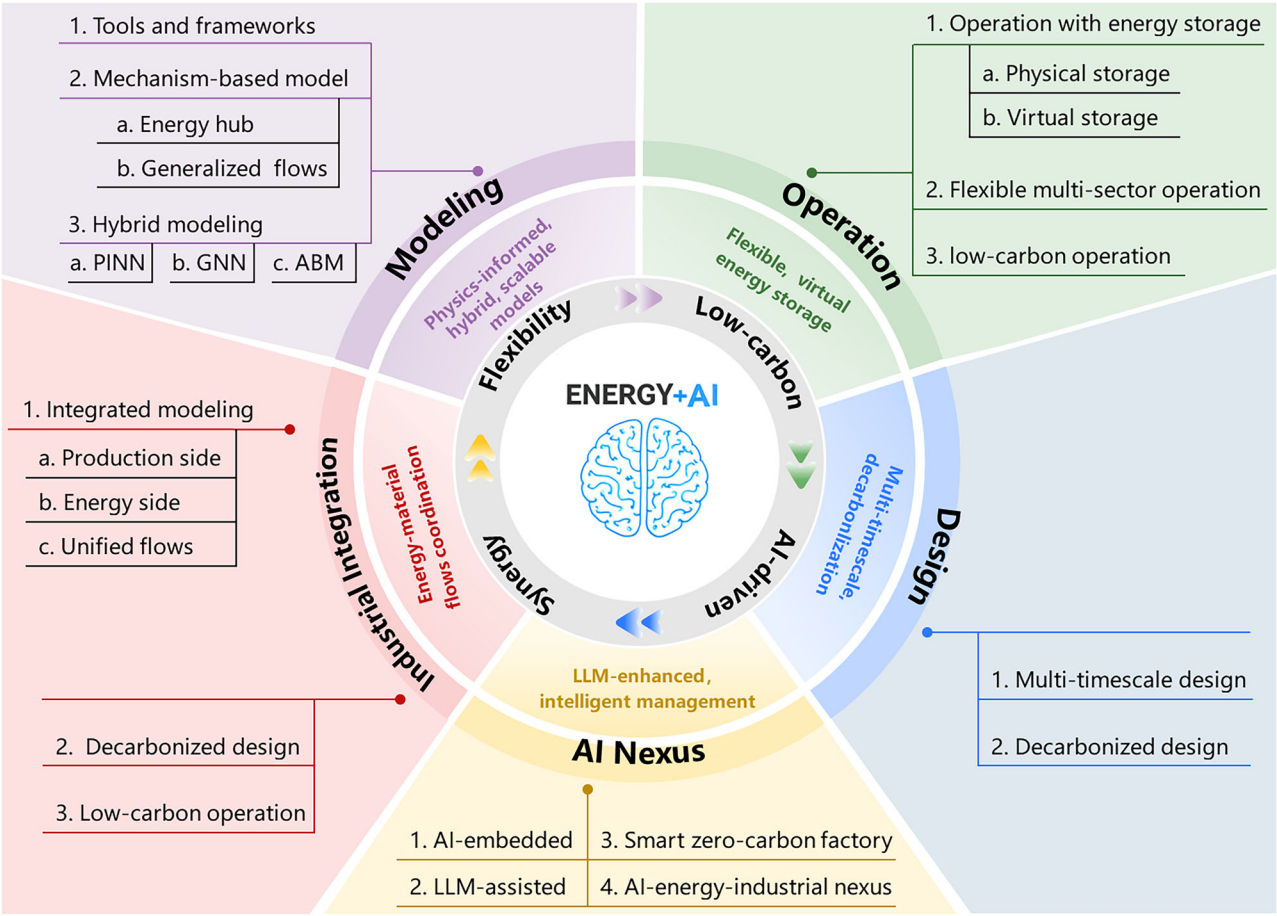

Fig. 3 | An illustration of the modeling, design, and operation framework for industrial integrated energy systems under the new paradigm of Energy and AI. The diagram depicts the framework through five interconnected components: (1) Modeling; (2) Operation; (3)Design; (4) Industrial integration; (5) AI Nexus.

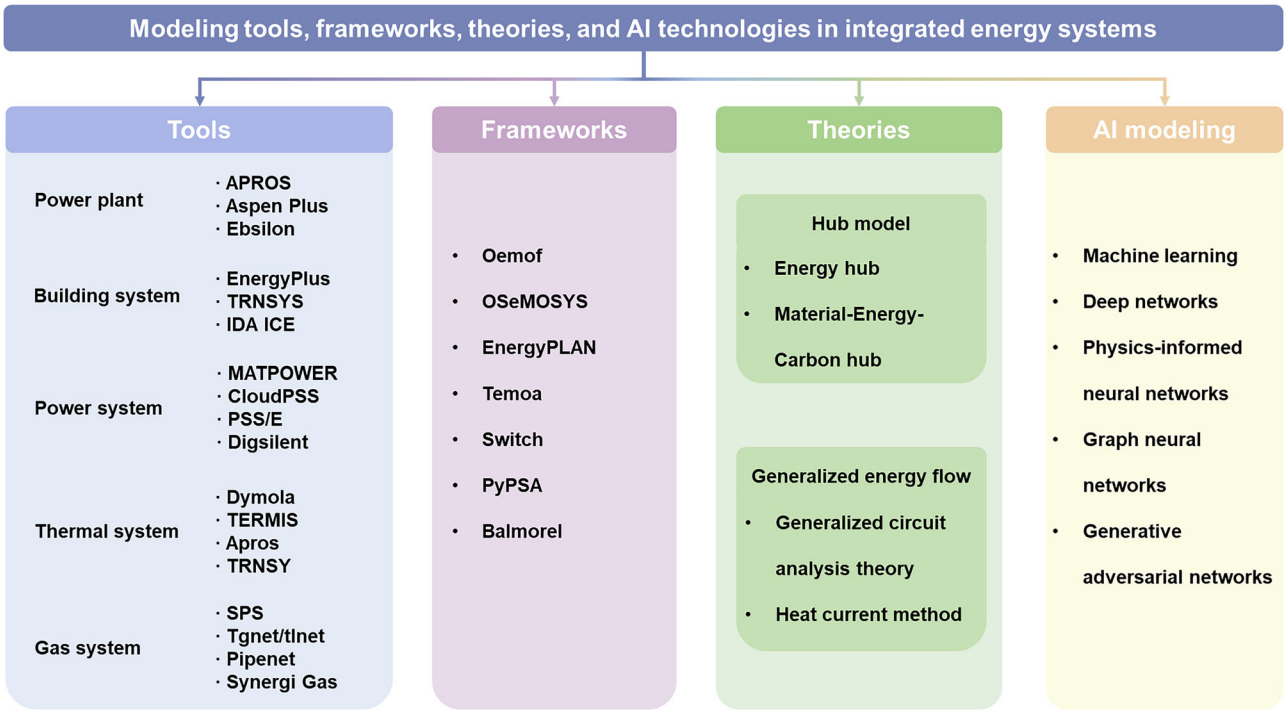

Fig. 4 | Overview of modeling tools, frameworks, theories, and technologies in integrated energy systems. The diagram categorizes the key components for analyzing integrated energy systems into four pillars: (1) Tools; (2) Frameworks; (3) Theorie; (4) AI Modeling.

Meanwhile, many open-source frameworks are available, such as Oemof[50], OSeMOSYS[51], EnergyPLAN[52], and TEMOA[53]. These frameworks facilitate the modeling and optimization of diverse energy systems, with some support for addressing features like uncertainty, nonlinearity, multi-temporal and spatial dimensions, and various energy carriers. Berendes et al.[54] proposed an ESMUT method and evaluated five major open-source energy system modeling frameworks. Groissböck et al.[55] evaluated 31 energy modeling tools, revealing that Switch, TEMOA, OSeMOSYS, and pyPSA excelled. Furthermore, many current open-source tools now match or exceed the capabilities of certain closed-source options. Nonetheless, these modeling tools and frameworks still face challenges in handling more personalized deployment scenarios and requirements. Additionally, existing modeling tools often encounter heterogeneous interface data, slow interaction times, and mismatched time steps when performing cross-scale co-simulation through software. Furthermore, most tools have limited control-level functionalities. Additionally, the closed architecture of many commercial software packages restricts the deep integration of third-party algorithms developed in Python and MATLAB code, making collaborative simulation across platforms and languages challenging. While these tools and frameworks provide the practical means for IES analysis, their foundation lies in the mechanism-based modeling theories, such as energy hub (EH) modeling and generalized energy flow modeling (EFM). These theories define the intrinsic principles and dynamic characteristics of multi-energy flows and core devices.

Energy hub modeling

The EH model was first proposed by Geidl et al.[56]. The model is defined as a unit capable of converting, regulating, and storing multiple energy carriers, and provides a general framework for modeling IESs. Numerous studies have conducted modeling work based on EHs for IESs that incorporate energy storage[57], energy networks[58], demand response[59], and renewable energy[60]. Nozari et al.[61] further explored the dynamic operating characteristics of multi-energy storage systems and developed a dynamic energy storage hub model. Mohammadi et al.[62] reviewed various concepts and models used for EHs, noting that future EH models should concentrate more on combined heat and power systems, renewable energy utilization, and applications across different sectors, such as residential, commercial, industrial, and agricultural. Pazmiño-Arias et al.[63] introduced a scheduling model for an IES in the dairy sector, leveraging the EH concept along with renewable energy and ice storage solutions. To tackle energy conservation and carbon reduction issues in the iron and steel industry, Zhang et al.[64] developed the traditional EH into a material-energy-carbon hub, analyzing how material and energy flows affect carbon dioxide emissions during production. Lasemi et al.[65] reviewed smart EH modeling methods from design, planning, and operation perspectives. They emphasized renewable energy, energy storage, demand response, optimization, and the significance of uncertainty, noting that robust and stochastic optimization is the leading method in c modeling.

Generalized energy flow modeling

Generalized EFM is a hot topic due to its coverage of energy carriers, including electricity, heat, gas, and steam, and their dynamic interactions. As demands for real-time operational performance, flexibility, and safety increase, traditional static models become inadequate for multi-timescale dynamic analysis. Generalized EFM presents a new way to model the dynamics of multi-energy flow networks, integrating graph theory and matrix analysis to describe mixed differential-algebraic systems uniformly. Jin et al.[66] reviewed the development history and solution methods of IESs, spanning from steady-state to dynamic modeling. They provided a detailed and comprehensive overview of research that incorporates dynamic characteristics into the modeling, design, operation, and control of IESs. Research examples include Yang et al.[67], who developed a generalized circuit analysis theory for multi-energy networks using circuit analogy and the Laplace transform. This describes the relationship between heat sources and loads, and the dynamic characteristics of heat networks. The theory also applies to unified modeling of electricity-gas IESs[68]. Chen et al.[69] abstracted elements such as "resistance," "inductance," and "capacitance" in gas, water, and heat networks based on the circuit analogy method and unified the mathematical forms and solution methods of multi-energy networks using the Fourier transform and graph theory. Zhang et al.[70,71] analyzed the similarities and differences in the physical structure and transmission processes of hot water and power networks, discussing various time–frequency domain transformation methods for dynamic modeling of electricity-thermal IESs. Chen et al.[72,73] identified relationships among thermal system elements and their consistency with power systems regarding heat transfer driving force and impedance, proposing another unified energy transfer model for electricity-thermal IESs.

The core of the dynamic unified solution methods for energy flows, such as electricity, heat, and gas, involves linearizing the model and partially decoupling equations (e.g., neglecting temperature in natural gas transmission, assuming flow and heat transfer are separate in heating systems) around stable, reliable reference conditions. It finds analytical expressions for energy and mass transfer relationships between adjacent nodes in the frequency domain and combines these with the energy system topology. However, for large-scale, complex looped structures and multi-energy flow networks under extreme conditions, these models involve numerous spatiotemporal variables and complex coefficients, so issues of computational efficiency and accuracy remain. This has prompted the development of a physics-informed hybrid modeling approach that integrates various neural network technologies to overcome limitations in model scalability, computational efficiency, and adaptability and improve prediction accuracy.

Physics-informed hybrid modeling

IES modeling often encounters a lack of sufficient historical data for model calibration, hindering frequent analyses of variable operating conditions and real-time system control. Many refs. 74–77 indicate that data-driven methods like machine learning, deep networks, and reduced-order models excel in modeling and predicting IES loads, equipment, and networks. They offer better accuracy, adaptive adjustment, and faster computation than mechanism models. Current challenges with data-driven approaches include the need for vast training data to create accurate models and the absence of reliable AI explanation frameworks[78]. Li et al.[79] noted that AI performance in multi-scale engineering is often inadequate, necessitating urgent focus on the logic and architecture of data systems for a high-quality ecosystem. A common solution is to model considering both mechanisms and data fusion in complex energy scenario systems. Two typical methods are to combine the physical physics-informed neural network (PINN), graph neural network (GNN), with IES modeling (see Fig. 5). These methods have been applied in fields such as multi-physics simulation, design optimization, multi-energy flow calculation, and state estimation[80–83]. Figure 5a shows the structure of a PINN, where the neural network's loss function is augmented with physical laws to ensure its predictions are physically consistent. Similarly, Fig. 5b depicts a GNN architecture that processes a system's network topology and time-series data to capture complex spatio-temporal dynamics. Furthermore, Fig. 5c illustrates the classic application of PINN and GNN in energy modeling.

PINN was initially introduced by Raissi et al.[84]. PINN's primary advantage lies in providing a unified framework for solving both forward problems (i.e., solving PDEs given physical parameters) and especially inverse problems (i.e., inferring unknown physical parameters from observed data), and enhancing data efficiency by regularizing the solution space with physical laws. At present, PINNs are being widely applied in power systems. Huang et al.[85] offer a review of the latest research on PINNs for power system state estimation, dynamic analysis, optimal power flow, and anomaly detection. In building and thermal energy systems, Xiao et al.[86] introduced an optimization framework utilizing physics-consistent deep learning aimed at load forecasting, indoor climate prediction, and enhancing grid-interactive community indoor thermal comfort control in energy systems for buildings. Gokhale et al.[87] presented two novel physics-informed neural network architectures and incorporated prior information about

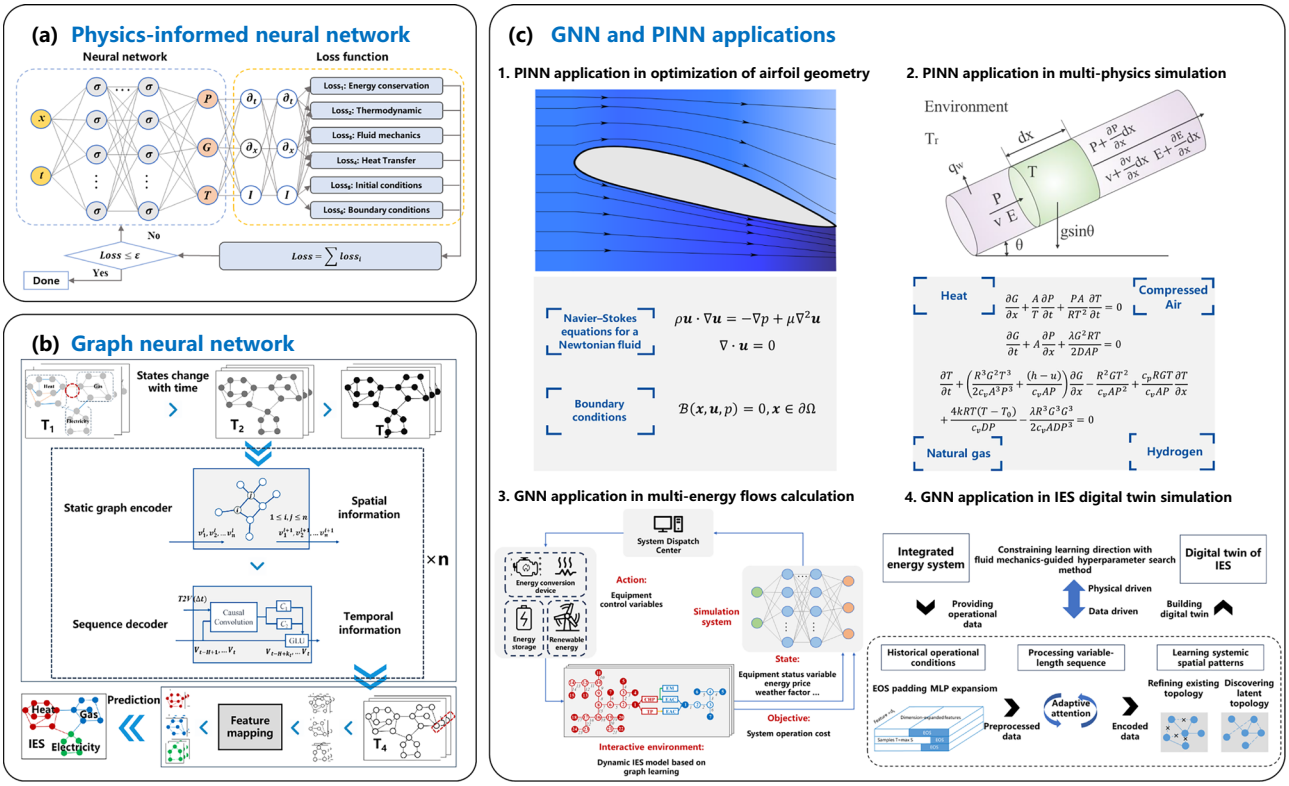

Fig. 5 | PINN and GNN for integrated energy system modeling. a Core architecture of a PINN. **b** A GNN architecture for spatio-temporal modeling. **c** Illustrative applications, including airfoil geometry optimization[295], multi-physics simulation, multi-energy flow calculation, and IES digital twin modeling.

building parameters into the neural network to enhance model interpretability. In geothermal energy systems, PINNs are employed for inverse modeling to forecast essential parameters, such as temperature, pressure, and permeability in geothermal wells[88]. However, current PINN-based modeling methods have not yet effectively addressed complex structures such as multi-node fluid networks and electricity-thermal-gas IESs. Additionally, the issues of training stability and generalization under extreme operating conditions remain unsolved.

GNN technique has also attracted significant attention. Unlike the PINNs paradigm, which depends on the explicit representation of partial differential equations, GNNs are skilled at adapting to multi-spatiotemporal scale prediction tasks[89] and can learn node-edge relationships directly from graph data structures[90]. This capability enables the effective capture of the diverse topological features and dynamic interaction mechanisms inherent within multi-energy flow networks. For instance, Yang et al.[91] integrated RC models and GNNs to develop dynamic thermal models for multiple regions. Their findings indicated that GNNs exhibit robust learning abilities and can handle up to 60% data loss. Yang et al.[92] introduced a physics-guided graph neural network (PG-GNN) approach to autonomously learn power grid topology, integrating it with transfer learning for real-time optimal power flow solutions. Boussaid et al.[93] introduced a physics-informed spatio-temporal convolutional GNN designed for quickly predicting transient physical fields in district heating networks. This innovative network achieves a 99% decrease in simulation time relative to conventional mechanistic models while maintaining high accuracy. However, GNN-based modeling relies heavily on large training datasets, may produce physically unreliable predictions, and lacks robustness and generalization capabilities. Future research should explore integration mechanisms and interpretability of graph dynamics with multi-physics conservation laws, while also addressing the generalization limitations of data-driven GNNs.

While the modeling methods mentioned earlier are predominantly focused on centralized IES, Agent-based modeling (ABM) offers a powerful framework for decentralized IES modeling. ABM primarily comprises distributed optimization algorithms (such as ADMM and consensus algorithms), game-theoretic approaches (including the Stackelberg game model), and multi-agent reinforcement learning techniques. A comparison of these three modeling types is presented in Table 2, highlighting the fundamental trade-offs involved in selecting these methods for various system objects and scenarios. At the application level, ABM technology has been employed in several crucial areas of integrated energy. To enhance system resilience and security, agents coordinate distributed energy resources to facilitate rapid recovery during extreme climate events and conditions. For instance, Dan et al.[94] developed an ABM platform to optimize the planning and layout of urban-level electric vehicle charging stations, demonstrating that multi-agent-based optimization strategies and vehicle-building-grid interactions can accelerate urban decarbonization and enhance the resilience of the energy system. Additionally, to promote low-carbon operations, the ABM integrates carbon trading mechanisms and emission constraints, providing a framework for point-to-point (P2P) energy and carbon trading. For example, Madler et al.[95] constructed an ABM to simulate the operational performance of urban micro-grids under different market conditions and explored the comprehensive performance of micro-grids in the economic, technical, and environmental dimensions under P2P energy trading. However, for ABM to large-scale IESs, breakthroughs still need to be made in algorithmic scalability and synchronization. Meanwhile, the decision-making of human-in-the-loop will also be a key research point for improving the trustworthiness of the model.

Industrial energy system modeling

With the above understanding of the evolution from traditional mechanism-based to advanced hybrid modeling techniques, the focus now turns to their application in one of the most complex and critical domains: the integrated modeling of industrial production and energy supply. Industrial energy systems are distinct from residential and commercial systems due to their high energy consumption and stringent safety requirements. To ensure a stable and efficient energy supply in industrial

Table 2 | Comparison of dominant methodologies for agent-based IES modeling

Method	Main Idea	Model requirement	Optimality guarantee	Limitations	Typical application	Ref.
Distributed optimization	Decomposing a global problem, iterative convergence to a system-wide optimum.	Requires an explicit mathematical model of the system.	Provable convergence, mathematical rigor.	Sensitive to communication failures, struggles with non-convexity.	Coordinated economic dispatch, voltage/frequency control.	286
Game theory	Modeling strategic interactions between rational, self-interested agents to find an equilibrium.	Less dependent on the physical model than optimization.	Guarantees convergence to an equilibrium, may not be globally optimal.	Assumes perfect rationality, hard to solve the equilibrium for many players.	P2P trading, electricity pricing, and demand response incentives.	287
Multi-agent reinforcement learning	Agents learn optimal policies through interaction with the environment.	Model-free, learns directly from data and interaction.	Generally finds near-optimal policies.	Sample inefficiency, scalability challenges.	Real-time adaptive control, resilience against extreme events.	288

settings, factories and industrial parks are often equipped with dedicated energy generation facilities. The integration of energy systems with industrial production has undergone a gradual and transformative evolution. In the early stages of industrialization, energy systems and industrial production operated independently, with excess energy and waste materials largely unutilized[96]. As energy management in industrial processes progressed, the development of CHP[97] marked a pivotal shift by facilitating the coupling of industrial production processes with steam and electricity, waste heat, and material recovery led to the initial integration of energy and production processes[98]. Research on this integration can be categorized into three types, its development in recent years can be seen in Fig. 6. The first focuses on the energy sector, where system scheduling is used to optimize industrial demand response (IDR)[99] through the flexibility of production processes. The second originates from production process management, in which multiple forms of energy are considered critical manufacturing resources, especially in energy-intensive industries[100] such as steel, chemicals, and machinery manufacturing[101]. As research advances, coupling models of industrial production and energy systems are evolving into a unified framework centered on energy and material flows, aiming to optimize both production processes and energy supply.

Modeling industrial load subsets for energy system analysis has traditionally involved treating these loads as components of overall energy demand. However, this approach often neglects the unique operational constraints and dynamics of production processes. With the continued evolution of electricity markets, there has been a growing focus on integrating IDR, particularly in industries such as cement and aluminum smelting[102]. For example, Mossie et al.[103] developed a mixed-integer linear programming (MILP) model to assess the demand response potential of waste heat power generation in cement plants. Similarly, Wang et al.[104] quantified the reducible load of ladle furnaces in the steel industry, considering peak-valley electricity pricing. Additionally, multi-time-scale modeling strategies for end-use industries have been explored, focusing on optimizing energy consumption across different time frames[105]. Industrial load subset modeling for IDR has traditionally concentrated on electricity. However, modeling gas and thermal energy consumption is becoming increasingly vital to the completeness of IES. To address this gap, researchers have investigated the interchange of energy between fuel and steam systems[106] within industrial energy models. As the demand for carbon management and green production grows, there is a need for more refined models of industrial loads, even in less energy-intensive sectors[107]. In response, Mu et al.[108] proposed a coupling representation in low-dimensional space for energy-consuming industries modeling, which meets the real-time control requirements of industrial operations.

Modeling energy resources in industry processes, particularly in energy-intensive industries, such as metallurgy and coal chemical production, is a dynamic system where material flows, energy flows, and waste streams interact and influence each other[101]. These industrial processes are basically continuous[109], with energy being a critical resource, or involve the processing and production of energy, such as in oil refining and hydrogen production. Early modeling research integrated the energy consumption module[110] into manufacturing system models, aiming to align energy management with production control and to develop energy efficiency assessment tools through data integration[111]. For example, Yuksek et al.[112] established a data collection and evaluation framework to quantify the embodied energy of product, enabling industries to measure production efficiency[113] and assess low-efficiency[114] processes.

In addition to energy data integration, energy-intensive industries have also pioneered the joint optimization of energy and material flows. For example, in the steel industry, the carbon-material-energy flows nexus in electric arc furnace processes has been studied[115], and Karimi-Zare et al.[116] developed a low-carbon optimization model that includes production planning and energy management, guiding low-carbon production in the steel industry. Zhang et al.[117] explored the coupling among carbon emissions, materials, and energy in fluid catalytic cracking units, while Xu et al.[109] conducted coupling optimization of material and steam systems for

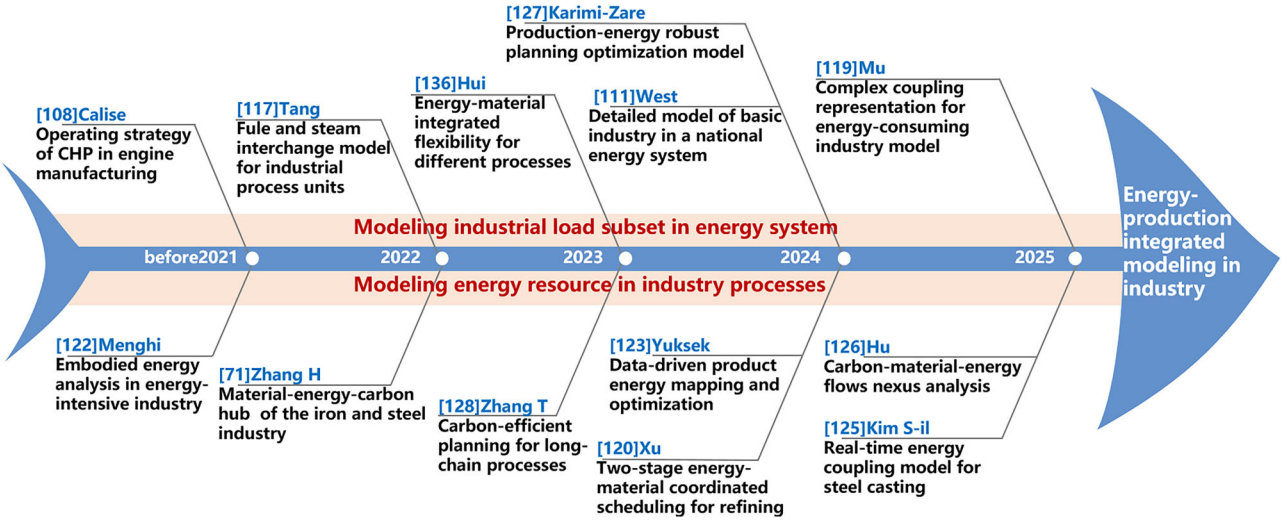

Fig. 6 | The recent research on energy-production integrated modeling in industry. This timeline highlights key research developments in modeling industrial load subsets and energy resources, tracking significant studies from before 2021 to 2025.

refineries. Collectively, these studies contribute to a deeper understanding of energy-material coupling from the perspective of industrial energy management.

Unified model for energy and material flows

Achieving holistic system optimization requires the joint scheduling of materials and energy, yet conventional fixed production plans often hinder such coordination. Recognizing this limitation, researchers have underscored the value of cross-sectoral interaction modeling[118] for both energy and material usage in manufacturing. Among modeling methods, graphical modeling for production processes offers distinct advantages in handling complex workflows, concurrent tasks, and resource management. For instance, Ding et al.[119] introduced a state-task network (STN), which represents the dependencies between tasks and states as a network, to model production as a combination of discrete and continuous variables. Similarly, Petri nets can represent the dynamic behaviors of processes, including concurrency and asynchrony[120], making them suitable for modeling production lines that integrate energy flows, material flows, and carbon flows[121].

On the other hand, several studies have designed a unified method to represent the conversion, storage, distribution, and consumption of energy and materials. To construct scalable models that couple energy and material flows, Li et al.[122,123] proposed a unified framework for industrial park planning, integrating material flows into EH models. Furthermore, Zhang et al.[64] introduced the integrated material-energy-carbon hub model, enabling precise energy efficiency optimization and carbon emission control at the specific production stage. Bao et al.[124,125] integrated an energy converter model with four different production tasks, exploring the integrated flexibility. Collectively, these advances establish a unified, scalable modeling framework capable of addressing diverse industrial systems and lay the groundwork for real-time, co-optimized management of materials and energy.

In conclusion, this section has examined the evolution of IES modeling, starting with a review of established software tools and frameworks and their inherent limitations in interoperability and integration. It then explored the core mechanism-based theories, such as EH and GEF models, which form the physical foundation of IES representation. The focus is now on the significant paradigm shift toward hybrid physics-informed models, especially PINN, GNN and ABM, which aim to address the limitations of purely physical or purely data-driven approaches. Ultimately, applying and adapting these advanced modeling techniques to the unique, tightly-coupled domain of industrial systems—where energy and material flows must be optimized together—represents the cutting edge of IES modeling and a vital step toward industrial decarbonization. To provide a clear comparative overview of the modeling approaches discussed, Table 3 summarizes their core principles, typical applications, strengths, and limitations.

Integrated energy system design and operation

Multi-timescale optimal

The optimal design and operation of IESs is a core scientific issue in building a sustainable energy system in the future. Current research commonly uses bi-level optimization models to couple the design and operation stages of IES design (or planning) methods. The design level selects the device and capacity configuration, using linear or nonlinear models and heuristic algorithms. The operation level employs MILP for scheduling optimization. Ding et al.[126] proposed a bi-level planning model integrating renewable energy and HVAC demand response. Results indicated that reducing electric chillers' capacity and increasing the planned capacity of ground source heat pumps and thermal storage lead to higher average exergy efficiency and flatter electricity/heat network curves. He et al.[127] focused on planning near-zero emission park IESs using a bi-level method to configure renewable energy capacity, power-to-gas, and carbon capture, considering the dynamic impacts of electricity, hydrogen, and carbon markets. Although existing research considers the impact of system operation at the planning level, future research needs to further consider the impact of electricity, carbon, and fuel market trading behaviors, as well as the impact of high renewable energy penetration. The planning process of IESs also involves multiple time scales, such as short-term, medium-term, and long-term. Cross-seasonal energy storage and seasonal multi-energy complementarity are widely used in medium and long-term system planning. For example, Zhang et al.[128] decoupled the annual net power curve into seasonal and hourly energy balance, using a robust optimization method at the short-term scale to address the uncertainty of wind and solar power output, and planning seasonal energy storage at the long-term scale. Zhou et al.[129] achieved seasonal imbalance of renewable energy in the Nordic region by configuring hydrogen storage. Zhang et al.[130] proposed a three-layer IES optimization framework. The long-term layer ensures seasonal source-load matching via a refined electricity-gas complementary model. The medium-term layer employs intraday energy transfer for extreme weather. The short-term layer focuses on real-time energy balance and economic benefits through intraday peak shaving. Lin et al.[131] proposed a quantitative method for evaluating multi-energy complementarity in long-term planning optimization in industrial park IESs, simulating load growth for the next 20 years. Results indicated that improved multi-energy complementarity enhances the system's long-term economy.

Table 3 | Comparison of different modeling methods for integrated energy systems

Category	Specific method	Core principle	Typical application	Advantages	Challenges
Commercial & open-source tools	Commercial software (Aspen Plus, EnergyPlus, APROS, etc.)	High-fidelity simulations based on physical equations for specific energy domains	Power plant thermodynamics, building energy analysis, network hydraulics	High precision, user-friendly interfaces	Closed architecture, expensive, slow for large systems
	Open-source modeling frameworks (Oemof, pyPSA, EnergyPLAN, etc.)	Modular, Python-based platforms for custom energy system models	Energy planning, multi-energy system optimization, algorithm research	Free and transparent, strong community	Steep learning curve, lacks polished GUIs, and usability varies
Mechanism-based theories	Energy hub model (EH)	A multi-input, multi-output model for energy conversion, storage, and distribution	District energy, multi-energy buildings, industrial hubs	Simple and scalable, easily extensible	Steady-state model, Oversimplifies conversion processes
	Generalized energy flow models (EFM)	Uses circuit theory analogies to model multi-energy networks dynamically	Electricity-gas-heat network analysis, stability, flexibility studies	Captures dynamic behavior, Integrates multi-energy flows	Computationally intensive, relies on linear assumptions
AI-based Hybrid Models	Physics-informed neural networks (PINN)	Neural networks that integrate physical laws into their learning process	State estimation and parameter identification in physical systems	Data-efficient, physically consistent solutions, Handles ill-posed problems	Struggles with complex networks, Training stability issues
	Graph neural networks (GNN)	Learns from graph-structured data, capturing network relationships	Spatio-temporal prediction, power grid optimization	Manages topology, faster, robust to missing data	Hard to interpret dynamics, Issues with unseen topologies

In contrast to design stages, the operation of IESs is more dynamic, influenced by various source-load uncertainties and multi-timescale characteristics. Traditional methods convert constrained physical problems into linear or mixed-integer linear programming problems, which are solved using mathematical solvers such as Gurobi, Cplex, and GLPK. These solvers incorporate various preprocessing, heuristic algorithms, and acceleration algorithms. Simultaneously, some researchers have developed general and extensible frameworks suitable for modeling and optimizing the scheduling of IESs. For example, Langiu et al.[132] proposed the COMANDO framework, which allows for component-based modeling of energy systems and supports dynamic and nonlinear characteristics, enabling flexible customization of design and operation optimization problems related to energy systems. In addition, heuristic algorithms such as particle swarm optimization (PSO)[133,134], genetic algorithm (GA)[135], and multi-objective evolutionary algorithm (MOEA)[136] are also widely used to solve multi-objective optimization problems in IESs. Research commonly combines mathematical programming, heuristic algorithms, stochastic or robust optimization[137,138], and model predictive control[139,140] to handle multi-timescale problems. This is achieved by coordinating operations in day-ahead, intraday, and real-time stages, which gradually reduces the impact of uncertainties and optimizes multiple objectives, including system economy, carbon emissions, and flexibility. Li et al.[141] introduced a coordinated multi-timescale optimization framework for IESs. This framework focuses on optimizing scheduling at a regional level for the day-ahead stage, facilitating rolling optimization of device outputs within individual IESs during the intraday stage, and adjusting device outputs in real-time. Wang et al.[142] proposed a multi-timescale optimization model for a park IES that accounts for demand responses and source-load uncertainties in the day-ahead stage. A rolling model was established to minimize operating costs in the intraday and real-time stages. Wang et al.[143] proposed a multi-timescale scheduling optimization strategy for seasonal hydrogen utilization, considering renewable energy and load variations in day-ahead scheduling. It utilizes rolling optimization for intraday scheduling to tackle forecast errors from wind and photovoltaic fluctuations and a chance-constrained method for real-time scheduling to ensure short-term supply–demand balance. Results indicated a 12% increase in renewable energy consumption. The optimization objectives in the research are often fixed, ignoring dynamic changes caused by multi-timescale variations. Dynamic multi-objective optimization poses a challenge for complex time-varying scenarios, where the objective function, constraints, and feasible solution space can evolve in real-time with changes in the environment or system state. Traditional algorithms struggle to balance global search efficiency and dynamic tracking, suggesting that AI algorithms could be a breakthrough in the future. Although the theoretical models for the design and operation of IESs are becoming increasingly sophisticated, there is still a gap between research and practice. Table 4 summarizes six recent and upcoming international IES case studies. A primary conclusion from these examples is that successful IES implementation is not based on a single template, but rather on tailored technological solutions designed for specific environments. These cases highlight a clear global trend towards more intelligent and specialized energy management, moving beyond traditional designs to incorporate smart controls, novel hardware, and sophisticated planning methods to achieve decarbonization.

Physical and virtual energy storage

In recent years, in-depth research has shown that operating IESs should consider fluid networks' dynamic inertia alongside energy storage devices' control features, ensuring accurate scheduling and control. Shirizadeh et al.[144] indicate that choosing various time scales for scheduling instructions significantly influences the optimal scheduling strategy in IESs. Additionally, the scheduling of steady-state models fails to capture the dynamic effects of multi-energy coupling amidst uncertainty and diverse spatio-temporal characteristics. The introduction of new physical energy storage devices such as compressed air energy storage[145], molten salt energy storage[146], flywheel energy storage[147], and supercapacitors[148], as well as

Table 4 | Typical design and operation cases of integrated energy systems

Project	Year	System types	Energy forms	Key technologies
Germany (EUREF-Campus)[289]	2023	5.5-ha urban quarter with over 150 enterprises	Electricity (solar, wind), heat (geothermal, waste heat), biomethane, cooling	Comprehensive smart energy management system for centralized and precise energy control
Chile (Aysén SMA)[290]	2023	Isolated regional electricity systems in the developing world	Electricity (diesel generation, hydro, wind), heat, hydrogen, firewood	Stochastic planning to capture uncertainty for decarbonization and environmental remediation
China (Muse-Ruili Industrial Park)[291]	2023	A cross-border China–Myanmar industrial park with abundant renewable resources	Electricity (solar, wind, hydro), heat (geothermal, steam), cooling, gas	Steam Carnot battery designed to enhance electro-thermal complementarity and meet industrial energy needs
America (Arizona Community)[292]	2024	Smart district with renewable energy and carbon capture technology	Electricity (solar, wind), cooling, heat, gas	Integration of modular energy management, incorporating AI-based time-series prediction and autonomous scheduling
Netherlands (Leeuwarden)[293]	2025	Regional business and shopping park	Electricity (solar, wind), heat (waste heat), gas	Incorporating energy planning into urban development with a dedicated network of public and private stakeholders
Australia (Offshore Renewable Energy Zone)[294]	2025	Offshore energy systems for industrial, transportation, and residential usage	Electricity (solar, wind), hydrogen	Large-scale battery energy storage systems and smart power management strategy

virtual energy storage in fluid networks, enables energy matching to achieve energy–power decoupling and time–space decoupling gradually. Due to their varying scales of inertial delay, the traditional rigid real-time balance of energy can shift towards flexible non-aligned energy matching.

In terms of hybrid physical energy storage, the structure of combining batteries with other forms of energy storage[149] is the most effective. For instance, Wang et al.[150] analyzed the high-frequency and low-frequency fluctuations of renewable energy using the wavelet transform. They employed a hybrid energy storage system combining batteries and supercapacitors to engage in demand response. This approach increased the renewable energy utilization rate by 16%, while the hybrid system extended equipment lifespan by 75%. Zhu et al.[151] developed a novel IES that couples electrical and thermal energy storage with an Organic Rankine cycle (ORC), enabling economical operation through flexible hybrid energy storage and waste heat recovery. Considering the high investment and construction costs of energy storage equipment, some refs. 152,153 have proposed strategies for the utilization of shared energy storage. These strategies aim to further tap the potential of energy storage and improve economic efficiency, but the business model and pricing mechanism still require further research. Although physical energy storage technologies are powerful, their high investment costs have always been a key factor restricting the economy of IES. Researchers have turned their attention to the virtual energy storage inside the system, which utilizes the dynamic characteristics of the existing energy network and load to simulate the effect of energy storage.

In terms of virtual energy storage in fluid networks, Wang et al.[154] proposed an optimal scheduling method for IESs, considering the flexibility of natural gas pipeline storage and the demand response capabilities of end-users. Wang et al.[155] developed a multi-frequency optimization model for IESs, integrating virtual energy storage of heating networks and achieving coordinated high and low-frequency regulation with high renewable energy penetration through electricity–thermal coordination. Hu et al.[156] proposed an operation optimization model for IESs considering the constraints of electricity, heating, and natural gas multi-networks. During winter, while centralized heating operates, buildings are partially integrated into the regulation of energy systems, acting as virtual energy storage. This is due to the heat storage capacity of their envelope structures[157,158] and users' variable thermal comfort preferences[159]. In industrial applications, steam networks involve strong coupling of multiple physical fields such as temperature, pressure, and phase change, and their complexity far exceeds that of conventional hot water and natural gas networks. To address the high demand for steam and compressed air in industrial parks, researchers have explored optimizing system operations by focusing on virtual energy storage within industrial pipelines for these resources. For instance, Zhong et al.[160] introduced a technology for piecewise linearization of physical properties. This innovation enables analytical modeling of the dynamic processes within steam heat networks and allows for the rapid resolution of the dynamic characteristics of these networks under minor temperature variations. Zhuang et al.[161] introduced the steam accumulator into industrial steam systems to mitigate renewable energy fluctuations, enabling complementary operation of electricity and steam through steam network simulation and IES scheduling. However, although a large amount of research has been conducted on the high-precision dynamic modeling of fluid networks, incorporating the dynamic constraints of fluid networks into the scheduling process of integrated energy systems will significantly increase the model dimension and solution difficulty. Therefore, most optimization scheduling studies considering virtual energy storage in fluid networks still adopt linearized or static models. At the same time, research on the energy storage characteristics of steam/compressed air pipeline networks is still in its initial stage, and the impact mechanism of key parameters such as phase change and pressure fluctuations on virtual energy storage capacity has not been quantified, lacking a universal operation optimization framework.

In summary, both physical and virtual energy storage are indispensable for enhancing the flexibility of IES. While physical energy storage relies on dedicated, capital-intensive equipment to store and release energy, virtual energy storage unlocks the inherent, often untapped, flexibility within

existing infrastructure like fluid networks and thermal loads. This creates a key trade-off: physical storage offers direct, high-performance, and dispatchable control, making it ideal for applications requiring rapid response and high power density. In contrast, virtual storage offers a low-cost solution by leveraging existing assets, but its capacity and response characteristics are indirect, highly dependent on the real-time state of the network, and pose significant modeling and control challenges. Therefore, a primary future research direction is the development of advanced co-optimization and control strategies that can synergistically manage both types of assets.

Flexible operation

As renewable energy sources rapidly grow, the need for enhanced flexibility and management in IESs has become exceptionally important. Research on flexibility is highlighted in the design and operation of energy systems, primarily focusing on power systems[162], district heating systems[163], and building energy systems[164], with a gradual shift towards IESs. Existing research defines and embodies flexibility as an aggregate concept, with optimization of resource allocation at its core, based on spatiotemporal uncertainties in both supply and demand. As IESs advance to a more refined stage of management and control, the decomposition and aggregation of multi-sector and multi-category flexibility resources become increasingly important. Flexibility will evolve from a perceptual concept to a theoretical framework, becoming integral to IESs. This section categorizes flexibility resources across various sectors (illustrated in Fig. 7) and outlines the operating strategies of these systems, considering demand response, distributed energy integration, and the coordination of multi-energy flows, among other flexibility resources.

Currently, flexibility on the supply side is primarily enhanced by retrofitting thermal power units[165], utilizing thermal-electric decoupling technology[166], and adopting deep peak shaving operation strategies[167]. The flexibility of both the network and the storage sides primarily arises from physical energy storage and the virtual energy storage of fluid networks. Research on enhancing the flexibility of fluid networks primarily combines their thermal inertia[168], variable flow rate[169], variable temperature[170], and topological structure[171]. Physical energy storage flexibility is achieved through charging and discharging strategies optimized with time-varying energy prices. Demand-side flexibility enhancement focuses on integrating distributed energy resources, user demand response, and grid interaction. For instance, Srithapon et al.[172] enhanced the system's operational flexibility and decreased the requirement for grid electricity by coordinating the scheduling of heat pumps, thermal storage systems, and electric vehicles. Nozarian et al.[173] proposed utilizing building clusters as energy hubs to enable coordinated energy efficiency and flexibility management. Zhou et al.[174] proposed generalized flexibility indicators to explore the flexibilities in building energy systems and discussed methods to enhance system flexibility under various control strategies. Some studies[175,176] have also integrated the building and heating sectors with the transportation sector by using power-to-heat and power-to-vehicle technologies to achieve flexible energy conversion and value flow. For instance, Zhou et al.[177] conducted a review on the flexible and collaborative optimization of elements such as renewable energy, hydrogen energy, energy storage, and the spatio-temporal sharing modes of land and air transportation in the airport energy ecosystem. Researchers have also studied flexibility assessment in energy systems. Xu et al.[178] used a mechanism-data-driven method to approximate the flexibility domain boundaries of steam systems under different operating conditions. Tie et al.[179] suggested applying a multi-energy flow flexible operation domain to clearly illustrate multi-dimensional operational and safety constraints. They calculated the volume of this flexible operation domain to define the flexibility of the integrated energy system. The research shows that coordinated operation of energy subsystems offers flexible services, and energy equipment in a multi-level system can also provide flexibility. However, current quantification models for flexibility across multi-energy and multi-sector systems face a trade-off between accuracy and scalability. Future research should focus on hybrid methods, such as developing adaptive model simplification techniques or using machine learning to train surrogate models that can quickly and accurately replace complex flexibility domain calculations, aiming to find a better balance between precision and computational speed.

For the operational optimization of IESs with multiple flexibility resources, Liu et al.[180] proposed a two-stage flexible optimization model for source-load-storage in IESs, considering resources such as CHP units, energy storage, and distributed wind and solar units. Karimi et al.[181] introduced a stochastic framework for the operational scheduling of a renewable energy-based IES, considering total generation costs, generation flexibility, and demand-side flexibility. The optimized strategy enhanced the generation flexibility index and the thermal generation flexibility index by 22.98% and 34.64%, respectively. Gao et al.[182] considered the impact of electricity market flexibility on the operation of IESs, simulating the complex interactions between the bidding and scheduling decisions of IESs. The results indicated that flexibility technologies affect the economics of electricity market participants and lead to the accelerated retirement of low-flexibility equipment. Ma et al.[183] addressed nonlinear energy conversion and heat transfer constraints in combined heat and power systems, proposing a two-stage thermal-electricity dispatch scheme for optimizing energy supply

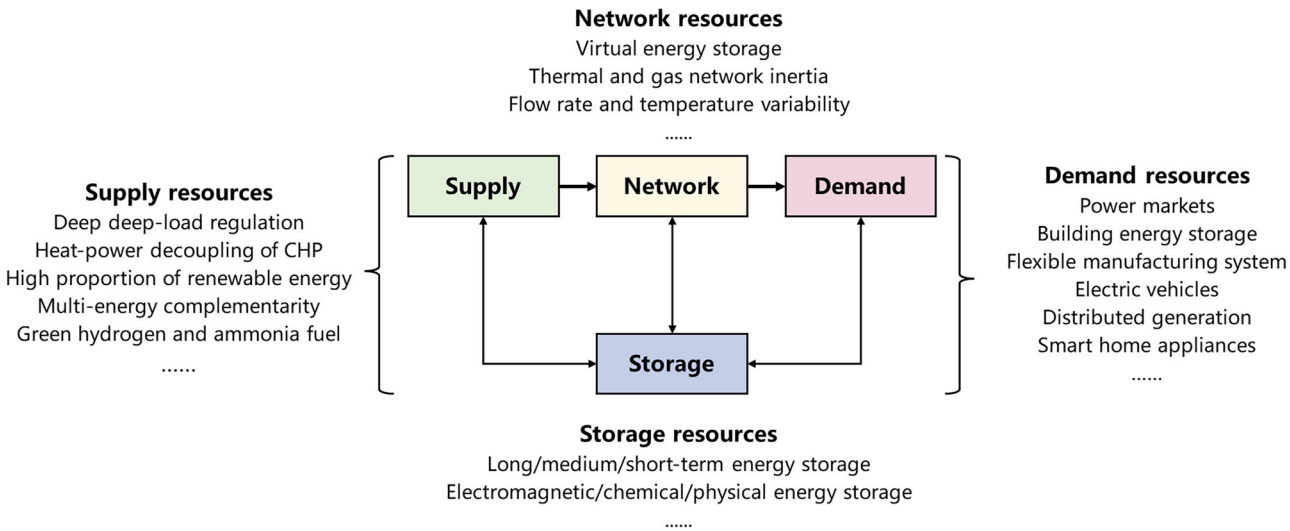

Fig. 7 | Source-network-load-storage flexibility in multiple sectors. This diagram illustrates the interactions between supply, network, demand, and storage resources, highlighting their flexibility in various sectors such as energy storage, power markets, and distributed generation.

flexibility and conversion efficiency. The results showed an 11.65% reduction in heat consumption compared to traditional methods. However, partial conflicts exist between flexibility enhancement and the economic and low-carbon goals of IESs. Focusing solely on flexibility optimization may hurt operational performance or raise carbon emissions. Future research must explore the relationships between flexibility and multiple optimization objectives, decoupling the inverse relationship by altering system configuration and operations for coordinated multi-objective optimization.

Decarbonized design and operation

Carbon emissions in industrial production originate from four primary sources: product design, resource control, the production process, and end-of-life disposal[184]. In response to increasingly stringent carbon regulations, research on industrial energy systems has coalesced around two complementary decarbonization pathways: embedding low-carbon technologies and optimized layouts at the design stage to reduce lifecycle emissions, and integrating carbon tracing, emissions constraints, and flexible control strategies into operation.

As for industrial IES design, an effective way must address each of these stages to achieve meaningful decarbonization. Early efforts emphasized optimized plant layouts and fuel substitution. Recent work has advanced toward multi-technology bundles tailored to specific sectors. One line of research is dedicated to integrating various low-carbon technologies, such as carbon capture, hydrogen, and heat pumps, with local renewable energy sources[101,185–187]. Recent work has advanced toward multi-technology bundles tailored to specific sectors. In heavy industries, Chen et al.[188] introduced a three-tier hybrid carbon-hydrogen process to reduce steel-plant emissions, while Bararzadeh et al.[189] studied the potential for producing low-carbon hydrogen and electricity in the steel manufacturing process. Wang et al.[190] evaluated the integration of photovoltaic power with steel production, proposing optimal energy substitution solutions suitable for different regions. Similarly, in light industries[191], coal substitution with green hydrogen has been shown to be feasible in plastics, chemicals, and fertilizers. Jost et al.[192] analyzed the low-carbon technologies implemented in flat glass production using LCA. By integrating biomass utilization and carbon capture technologies, Meys et al.[193] achieved net-zero emissions in the plastics industry. Zanon-Zotin et al.[194] explored sustainable development solutions for the chemical industry. Importantly, these low-carbon technologies must align with the characteristics of each process and local renewable sources[195]. Furthermore, from a broader global industrial perspective, strategic adjustments to production locations and energy supply structures can also enable decarbonization across entire production value chains[196].

As for industrial IES operation, carbon emission targets have been incorporated into industrial IES operations to promote clean production and sustainability. Table 5 shows an overview of optimization research in industrial IESs. Earlier operational strategies mainly focused on resource efficiency and environmental performance[197], while recent studies include specific carbon emission constraints, such as carbon markets, allocation, and footprints[198]. For example, Mengesha et al.[199] proposed that carbon pricing could promote green growth for energy producers. Fisco-Compte et al[200] incorporated carbon trading policy to achieve green production scheduling in manufacturing. In particular, Li et al.[201] contributed by focusing on the joint optimization of total carbon emissions and emission intensity in electrolytic aluminum production, utilizing an STN model. In multi-energy coupled production processes, low-carbon operations must balance the end-use energy loads of different energy sources. To address this challenge, Ma et al.[202] developed a linearization operation model with a nodal energy-carbon price, and Wu et al.[203] optimized the scheduling of the steel industry using wind-hydrogen power. Building on these system-level studies, Zhang et al.[204] recently used an integrated energy-material-carbon flows model in discrete manufacturing, achieving flexible production and precise footprint tracing.

At a more detailed level of production, research has focused on optimizing job shop scheduling and machine operations with a dual emphasis on energy efficiency and low-carbon performance. For instance, Fang et al.[205] pioneered the integration of energy consumption and carbon footprint into the optimization of dual-device scheduling in flow shop production systems. Piroozfard et al.[206] innovatively minimized both the total footprint and the total late work criterion in flexible job shop scheduling. Building on these advancements, Wang et al.[207] introduced an enhanced memetic algorithm that accounts for machine restarts, thereby improving adaptability in dynamic processes. Furthermore, logistics[208] and guided vehicles[209] in a job shop have coordinated these systems with energy scheduling, ensuring that material handling and energy management remain closely aligned.

Energy and AI nexus in future integrated energy system

AI-embedded operation

Recently, AI algorithms like deep learning (DL), generative adversarial networks (GAN), and reinforcement learning (RL) have significantly advanced IES optimization. Thus, it's essential to examine research on optimizing these systems beyond mathematical programming and heuristic algorithms. Existing research applies GANs to address scenario generation under source-load uncertainty. Huang et al.[210] analyzed correlations between source-load uncertainty variables and created multiple operating scenarios using GANs, enabling stochastic scheduling that addresses this uncertainty. Li et al.[211] used GANs and clustering algorithms to create scenarios and designed a metaheuristic method for scheduling multi-community IESs. Overall, the studies are limited to using AI algorithms for system boundary conditions and scenarios, without effectively solving operation scheduling problems in IESs. AI algorithms have advanced in unit commitment (UC) optimization to address power system scheduling challenges. To address this, Zhang et al.[212] proposed a frequency-constrained UC framework for high renewable energy scenarios, reformulating the deep neural network as mixed-integer linear constraints in a standard unit commitment model. Ramesh et al.[213] introduced a machine learning model that combines GNNs and LSTMs, reducing the computational complexity of security-constrained UC while maintaining solution quality by predicting generator schedules and transmission lines. Currently, the area closest to IES scheduling and control is RL[214], particularly deep reinforcement learning (DRL)[215]. RL is an approach that learns optimal strategies by interacting with the environment. It models scheduling issues as Markov decision processes (MDPs) and employs deep neural networks to approximate policies or value functions, allowing for adaptive optimization in complex, dynamic settings. IESs have a more complex state space compared to scenarios like building energy management[216], microgrid control[217], and the scheduling of electric vehicle clusters[218]. Perera et al.[219] reviewed the application of RL in energy systems, noting that RL methods are more competitive for large decision space problems than other technologies, particularly given uncertainties in renewable energy and energy markets and expanded energy system boundaries involving sector coupling. For instance, Yang et al.[220] enhanced the quality and learning efficiency of scheduling strategies using an improved deep deterministic policy gradient (DDPG) algorithm, addressing the optimal scheduling challenge of IESs while factoring in energy demand and renewable energy output uncertainties. Zhang et al.[221] employed the soft actor-critic (SAC) algorithm to tackle the multi-objective optimization challenge of balancing economy and supply reliability in electricity–thermal-gas IESs. The findings revealed that the benefits gained through the SAC-based DRL approach were 21.66% superior to those obtained from the PSO algorithm.

However, although model-free RL avoids complex mechanistic modeling, it still requires significant historical training data. Acquiring high-quality data in energy systems is often limited, especially for new IESs during the initial stages. RL needs extensive offline training to optimize decision strategies. The "black-box" approach lacks transparency and requires high-performance computing. Future research should investigate hybrid modeling, transfer learning, and explainable AI to improve DRL's decision-making in IESs. Current studies primarily focus on AI integration to optimize physical and mathematical models, while there is less attention on

Table 5 | Overview of optimization research in industrial integrated energy systems

Ref.	System	Uncertainty	Flexibility	Multi-timescale	Dynamic	Energy production coupling	Solution	Summary
64	Electricity, heat, compressed air	×	×	×	√	√	LP	Carbon flow tracing of the steel industry
104	Electricity, heat	×	√	×	√	√	MILP	Flexibility evaluation of the steel refining process
109	Electricity, heat, hydrogen	√	×	×	√	√	MINLP & Stochastic programming	Material–energy coupling optimization for refineries
116	Electricity, heat	√	×	√	×	√	Robust optimization	Production plan and energy management in the steel industry
202	Electricity, gas, cooling, heat	×	×	×	√	×	MILP	Low-carbon economic operation of energy hubs
201	Electricity, heat	√	×	×	√	√	MILFP using the Charnes–Cooper transformation	Emission intensity optimization for electrolytic aluminum based on STN
200	Electricity, gas	×	√	√	√	√	MILP	Production scheduling considering carbon policy
103	Electricity, heat	×	√	×	×	√	MILP	Demand response in the cement industry
207	Electricity	×	√	√	√	√	Enhanced memetic algorithm	Flexible job shop scheduling considering machine restart
209	Electricity	×	√	×	√	√	Improved evolutionary algorithm	Low-carbon flexible job shop scheduling
105	Electricity, heat, cooling	√	√	√	√	×	KKT and binary expansion method	Multi-time scale demand response for manufacturing
203	Electricity, gas, hydrogen	√	√	×	√	√	Lagrangian benders decomposition	Low-carbon dispatch in the steel industry, coupled with hydrogen
204	Electricity, heat, gas, compressed air	×	√	×	√	√	Relaxation and heuristic algorithms	Carbon flow tracing and coordinated scheduling of energy-material flows

innovative optimization algorithm designs. Current research frequently employs machine learning to adaptively modify the hyperparameters of optimization algorithms[222] or adjust the weights of various optimization objectives within the system[223]. However, these methods still find enhancing the algorithms' generalization performance challenging. Tang et al.[224] presented a comprehensive review of research on automated optimization algorithm design, recommending the use of automated machine learning methods to establish the mapping between target problem classes and solvers. This concept facilitates generative design and learning transfer of intelligent optimization algorithms for IESs by creating a problem feature space that includes typical energy scenarios and requirements, enabling automatic generation of tailored optimization algorithms for specific problem characteristics.

LLM assisted operation

LLMs have demonstrated significant potential in recent years beyond text[225], image[226], and video[227] domains. The energy, power, and building energy system sectors, characterized by high complexity and multiple constraints, are actively exploring the applications of LLMs in areas such as prediction, modeling, decision-making, and control. Research in power systems indicates the feasibility of using LLMs to solve optimal power flow[228], wind/solar power prediction[229], and state estimation[230]. Majumder et al.[231] examined the efficacy of LLMs in power system knowledge question answering, data analysis, and fault diagnosis. Nonetheless, they identified challenges, including the insufficiency of training datasets within certain domains and the lack of security and physical interpretability. Similar emerging results exist in building energy systems. Liang et al.[232] proposed EnergyGPT, which fine-tunes a pre-trained GPT-2 model to achieve high-accuracy multi-energy load forecasting. Jiang et al.[233] proposed the integration of LLMs with EnergyPlus to facilitate automated modeling. This approach utilizes fine-tuned LLMs to generate building models from natural language descriptions, thereby significantly reducing the modeling workload by 95% while maintaining high accuracy. Li et al.[234] developed an LLM-based query paradigm that combines model fine-tuning with a knowledge graph and retrieval-augmented generation to process complex operation and maintenance data of building systems. Overall, the advantages presented by large language models in the domains of text and data analysis empower them to manage equipment diagnosis effectively[235], as well as various prediction tasks[236,237]. Nevertheless, decision-making in energy systems utilizing LLMs encounters security and trust challenges[238]. The statistical findings presented by Zhang et al.[239] further corroborate that system operation and control remain the least explored domain within the context of LLMs, largely due to the optimality and physical constraints associated with the intricacies of these systems.

Some scholars have proposed integrating LLMs into agent-based modeling and simulation[240]. Multi-agent systems possess the capability to manage complex decision-making tasks through the interactions among various agents. Consequently, they hold considerable application potential within energy systems that routinely engage with operational entities, including electricity and carbon trading markets, as well as virtual power plants[241,242]. However, it is important to note that multi-agent systems based on LLMs rely on text and data-driven probabilistic reasoning, making it difficult to satisfy the physical equations and constraints of IESs strictly. To bridge this gap, Yang et al.[243] proposed an innovative LLM plus RL approach, which utilizes an LLM to generate and iteratively optimize penalty functions for an RL agent, thereby achieving safer energy management of active distribution networks while reducing reliance on human domain knowledge. Simultaneously, LLMs' current long logical reasoning time poses challenges for supporting dynamic regulation and ensuring the safe operation of energy systems under extreme weather impacts[244]. Significant bottlenecks remain in developing energy-LLMs, particularly at the process and algorithm levels. On one hand, collaboration between enterprises and academic institutions is essential to establish a transparent and shared, high-quality simulation data foundation for the training of LLMs, enabling self-learning and the autonomous generation of operational regulation

strategies. On the other hand, top-level design of energy-LLMs requires experts in materials, equipment, thermodynamic cycles, and subsystems to develop specialized modeling, simulation, and optimization tools, integrating multi-scale joint tools for global optimized design and operation. Concurrently, Jiang et al.[245] emphasized that the extensive training, fine-tuning, and updating of LLMs may contribute to significant carbon emissions and electricity consumption. Consequently, while employing LLMs to assist individuals in various tasks, it is imperative to consider their impact on global energy resources and the environment.

To conceptualize how these components might be integrated, Fig. 8 presents a potential layered architecture for an LLM-powered management and control system. This framework illustrates a bottom-up data flow, starting from a physical layer containing the IES assets, moving through a perception layer for multi-modal data collection, and feeding into a central intelligence layer. Here, the LLM acts as the core reasoning engine, integrating with digital twins and multi-agent systems to perform tasks like prediction, optimization, and decision-making before delivering actionable insights through an application layer. Such an architecture provides a conceptual model for the top-level design required to harness the potential of LLMs in future industrial IESs.

Zero-carbon smart factory

Industry 4.0 is built on the automation of machines and production systems, emphasizing the realization of intelligent processes through technologies such as smart devices, the Internet of Things (IoT), big data analytics, and cloud computing. Initially, driven by advancements in IoT[246], smart factories utilized real-time data monitoring of smart machines, logistics, and manufacturing processes to enhance energy management, leading to a closer integration between energy consumption and production activities. In this context, the accuracy of data and refined energy management are particularly critical. In practice, Guo et al.[10] employed high-resolution databases to systematically manage energy infrastructure. Tibrewal et al.[247], through a combination of survey data and remote sensing, identified significant disparities in energy use within brick production processes and proposed improvements to enhance energy efficiency.

Moreover, smart factories are increasingly adopting more sophisticated sensors and real-time analytics throughout the production process. By integrating AI technologies, smart factories enable real-time monitoring[248] of energy and material flows, while also allowing data-driven operations to energy and production equipment[249]. For instance, Kanoun et al.[250] and Li et al.[251] investigated the application of energy-aware technologies based on

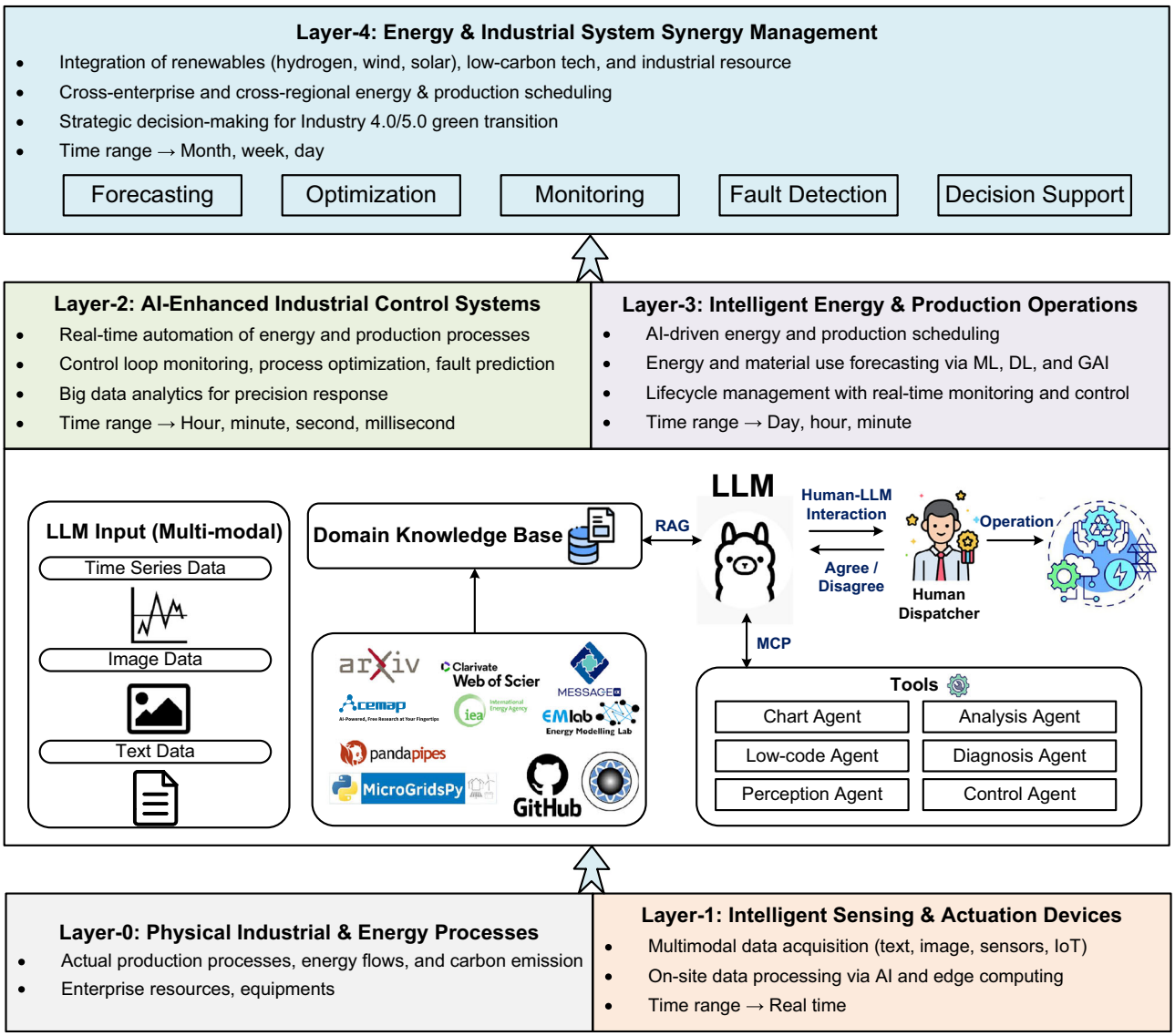

Fig. 8 | The architecture of an LLM-powered industrial integrated energy system management and control system. This diagram integrates five layers: the bottom layer includes physical industrial equipment and intelligent sensing, the middle layers involve AI-enhanced control systems and intelligent operations, while the top layer focuses on energy and industrial system synergy management.

wireless sensor networks (WSNs) to enhance factories' operational efficiency. Perera et al.[252] reviewed the AI-driven soft sensors in process industries, which can monitor energy-related parameters that are otherwise difficult to measure directly. Leveraging cyber-physical systems (CPS), Matsunaga et al.[253] explored real-time optimization of energy efficiency to improve smart manufacturing processes. Zhou et al.[254] further demonstrated how machine learning can analyze real-time data to forecast future energy demands.

With the rise of Industry 5.0, human–machine collaboration has emerged as a central theme. Alongside improving production efficiency, it emphasizes personalized manufacturing and sustainable development. The synergy among machine intelligence, digital twins, and augmented reality (AR) technologies has opened new pathways for energy conservation and carbon reduction[255]. Akundi et al.[256], for example, integrated AR and collaborative robotics to improve process flexibility, demonstrating how enhanced robot adaptability can simultaneously optimize energy management and reduce emissions. In parallel, AI-powered smart devices and networks offer more precise control over energy and carbon emissions[257]. Tao et al.[258] further explored the potential of digital twin technologies in industrial energy management, highlighting their capacity for real-time monitoring, prediction, and process optimization. Nevertheless, significant challenges remain in system integration, cost control, and workforce development, and addressing the short-term risk of increased greenhouse gas emissions during smart factory deployment remains an urgent engineering issue[259].

AI-energy-industrial nexus

With the development of renewable energy and the pursuit of carbon neutrality, industrial processes are increasingly integrating sustainable technologies and renewable sources such as hydrogen[260], while advancing next-generation intelligent production systems aligned with Industry 4.0 and 5.0 paradigms[261,262]. AI technologies not only support the storage, conversion, and management of sustainable energy, but also facilitate the integrated optimization of regional industrial resources, thereby promoting low-carbon and green regional transitions[263]. As shown in Fig. 9, the integration of AI and renewable energy is mutually reinforcing, and this multidisciplinary digital evolution optimizes the management of energy, production, and carbon, further accelerating industrial decarbonization. AI-driven industrial energy-carbon optimization utilizes technologies such as machine learning, intelligent algorithms, and generative artificial intelligence (GAI) to optimize industrial processes. A key step is the accurate prediction of energy consumption and material usage. Common techniques used include machine learning (such as support vector machines and decision trees), DL[264,265], and uncertainty modeling[266]. For example, Tan et al.[267] combined tree algorithms, recurrent neural networks (RNNs), and regression models to predict machine activity states, production capacity, and energy allocation. Other work has merged AI with big data analytics. Majeed et al.[268] developed a framework for sustainable additive manufacturing, while Liang et al.[269] combined AI optimization with data envelopment analysis (DEA) to guide targeted energy-saving and carbon-reduction measures. Collectively, these prediction and optimization methods enable real-time analysis and precise adjustment of energy management and operational strategies[270].

A key direction for future industrial decarbonization lies in the integration of AI with low-carbon energy and sustainable technologies. In the chemical industry, Mallapragada et al.[271] demonstrated how AI can optimize electricity conversion and distribution, significantly improving energy efficiency. Beyond electrification, AI-supported thermal process optimization offers further emission reduction potential[272]. And AI is expanding into more advanced sustainability technologies, such as carbon capture. Manikandan et al.[273] demonstrated improvements in capture efficiency and energy use. Increasingly, such approaches are being applied at system and regional scales. Xie et al.[274] proposed a comprehensive data-driven framework for low-carbon regional transitions to reconfigure industrial resources.

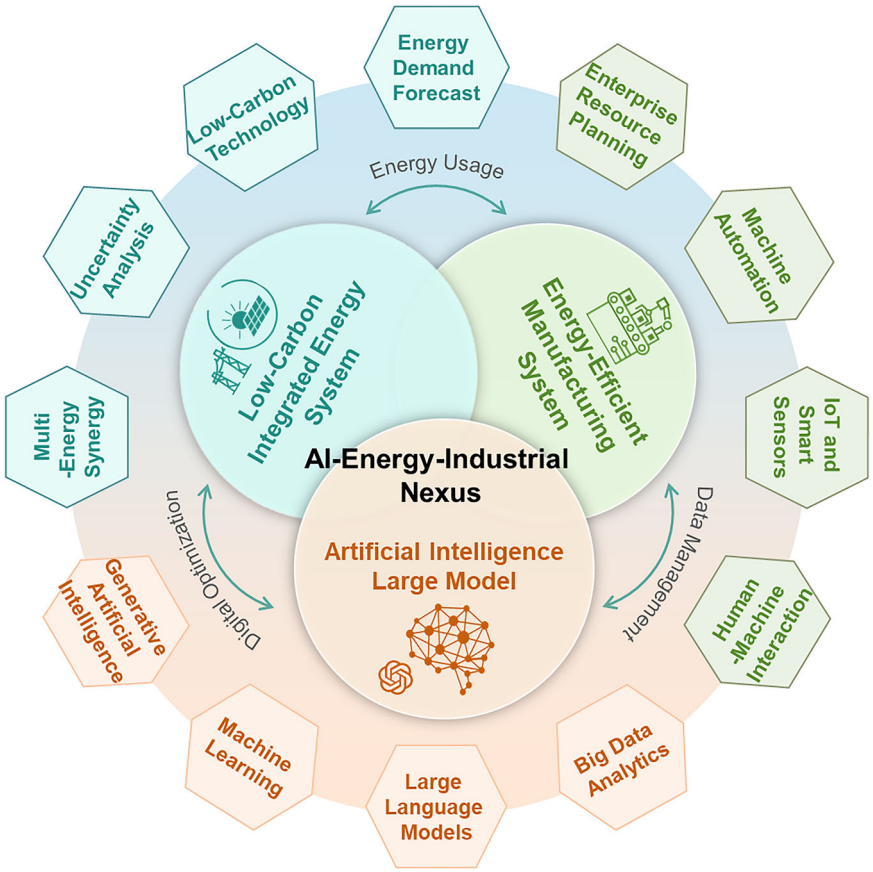

Fig. 9 | AI, energy and industry multidisciplinary nexus. This diagram illustrates the intersection of AI, energy systems, and industrial processes, highlighting low-carbon integrated energy systems and energy-efficient manufacturing, along with supporting AI technologies.

Complementing this, Chinnathai et al.[275] introduced an AI-supported life-cycle management model enabling real-time monitoring and adaptive control in energy-intensive sectors.

Meanwhile, rapid advances in LLM are opening new possibilities for scaled intelligent and low-carbon manufacturing[276]. By processing multi-modal data—such as text, images, and sensor inputs, LLM can support various industrial scenarios and complex tasks. One key application is predictive maintenance, where equipment data is analyzed to detect potential failures in advance[235], thereby minimizing unplanned downtime. LLM is beginning to play a role in optimizing operations within power systems[277], further demonstrating its versatility in industrial contexts. With strong digital infrastructure, abundant data resources, and an innovation-driven industrial base, China is well-positioned to lead this transformation[278]. For instance, Huawei's Pangu model targets sectors such as mining and railways, while Alibaba and 01.AI have co-founded the "Industrial Large Model Laboratory" to develop enterprise-focused LLM solutions. In summary, the AI-energy-industrial nexus represents a critical enabler of deep decarbonization, where advances in prediction, optimization, and control intersect with renewable integration and intelligent manufacturing, creating synergistic pathways toward zero-carbon industrial systems.

Conclusion

Despite significant advancements in the modeling, design, and operation of IESs, developing a smart operating system that fully integrates multi-energy flow theory with AI is still a promising area for future exploration. This review provides an overview and analysis of contemporary smart operation technologies for IESs from three viewpoints: modeling methods, operational strategies, and the nexus of AI, energy, and industry. In conclusion, the transition to clean and low-carbon energy, especially in the industrial sector, requires advanced modeling, design, and operational strategies for IESs. This review emphasizes the critical role of AI, ranging from improved modeling techniques and smart operations to the incorporation of LLMs for enhanced decision-making. Addressing the inherent complexities and uncertainties of future IESs requires continued interdisciplinary research, focusing on the synergistic integration of AI with domain expertise to unlock the full potential of smart, efficient, and sustainable energy solutions in pursuing carbon neutrality. As we focus on innovative IESs aimed at achieving carbon neutrality, future research needs to advance interdisciplinary integration with AI and leverage AI to address the challenges of collaborative operation in ultra-large-scale complex systems.

To bridge the gap between research and industrial IES application, two key technical challenges must be addressed: (1) There is a challenge in aligning data-driven AI models with physical laws governing energy systems. Current AI often overlooks thermodynamic principles and system dynamics, leading to unsafe decisions. Developing hybrid models that incorporate these physical constraints is crucial. (2) Future IES must manage various objectives (cost, reliability, flexibility, emissions) under uncertainties from renewable energy and market interactions. Current systems lack the real-time adaptive capabilities needed for efficient operation. Transitioning to AI-driven adaptive control within physical constraints is essential for advancing smart energy systems.

Addressing these bottlenecks will require a multi-stage effort with deep interdisciplinary collaboration:

(1) The immediate focus should be on creating a cross-scale, physics-constrained hybrid modeling framework. Utilize AI to accelerate multi-scale simulations and potentially discover unknown thermodynamic and dynamic equations from data. This endeavor requires a close collaboration between thermal and power systems engineers, who offer domain expertise, and AI and computer scientists, who can construct the necessary hybrid model architectures.

(2) The next step is to achieve intelligent and adaptive operation. Develop AI agents that can manage multiple operational objectives and significant load variations with high computational efficiency. These systems must intelligently orchestrate the flexibility of diverse resources across the entire IES. This phase requires the inclusion of control theorists to ensure system stability, as well as economists to integrate market dynamics and complex decision-making processes.

(3) The ultimate goal is to create a multi-modal cognitive engine for holistic system management. Develop a "Language-Physics-Spatio" multi-modal LLM. This framework would integrate physical laws via symbolic equation encoding, analyze real-time geospatial and meteorological data, and support natural language interaction for dispatchers. This visionary step will require an unprecedented synergy between energy experts, AI researchers (in NLP and computer vision), and geospatial scientists to create a truly intelligent human-AI partnership for managing complex energy systems.

Based on our analysis, we propose the following directions for future guidance:

(1) Advancements in AI models require large, high-quality datasets. It is important to create standardized data-sharing platforms across energy subsystems (electricity, heat, and gas) and prioritize the development of digital twin technology. By creating a real-time digital mirror of the IES, we can simulate various extreme scenarios safely.

(2) Future AI models should move beyond "black-box" correlations to systems that respect thermodynamics, fluid dynamics, and device characteristics. Focusing on hybrid models like PINNs and advancing explainable AI will foster trust among operators by ensuring transparent and interpretable AI-driven decisions.

(3) Technological breakthroughs require a supportive market and policy environment for their successful implementation. Future research must integrate technological innovation with market mechanism design and the refinement of policies and regulations. For example, how can the flexibility, reliability, and environmental benefits enabled by AI-driven dispatch be accurately priced? How can new electricity, carbon, and ancillary service markets be designed to incentivize IES operators to adopt smarter and lower-carbon operational strategies?

(4) The goal is to empower, not replace, human experts. Future work should develop domain-specific LLMs for the energy sector. These models can interpret natural language commands, clarify complex events, and propose operational strategies, creating a synergistic system where human oversight complements AI's optimization capabilities.

Data availability

No datasets were generated or analyzed during the current study.

Received: 30 April 2025; Accepted: 17 September 2025;
Published online: 02 February 2026

References

1. UNFCCC. *Paris Agreement: Decision 1/CP.17-UNFCCC Document FCCC/CP/2015/L.9/Rev.1* http://unfccc.int/resource/docs/2015/cop21/eng/l09r01.pdf (2015).
2. World Meteorological Organization. *State of the Global Climate 2024* https://library.wmo.int/viewer/69455 (2025).
3. Van Soest, H. L., Den Elzen, M. G. J. & Van Vuuren, D. P. Net-zero emission targets for major emitting countries consistent with the Paris agreement. *Nat. Commun.* **12**, 2140 (2021).
4. Rogelj, J. et al. Paris Agreement climate proposals need a boost to keep warming well below 2 °C. *Nature* **534**, 631–639 (2016).
5. Chen, P. et al. The heterogeneous role of energy policies in the energy transition of Asia-Pacific emerging economies. *Nat. Energy* **7**, 588–596 (2022).
6. Wang, Y. et al. Accelerating the energy transition towards photovoltaic and wind in China. *Nature* **619**, 761–767 (2023).
7. Aust, V., Morais, A. I. & Pinto, I. How does foreign direct investment contribute to Sustainable Development Goals? Evidence from African countries. *J. Clean. Prod.* **245**, 118823 (2020).

8. Le Tellier, M., Berrah, L., Stutz, B., Audy, J.-F. & Barnabé, S. Towards sustainable business parks: a literature review and a systemic model. *J. Clean. Prod.* **216**, 129–138 (2019).
9. IEA. *Energy Efficiency 2023* https://www.iea.org/reports/energy-efficiency-2023 (2023).
10. Guo, Y., Tian, J. & Chen, L. Managing energy infrastructure to decarbonize industrial parks in China. *Nat. Commun.* **11**, 981 (2020).
11. NBS. *China Statistical Yearbook 2024* https://www.stats.gov.cn/sj/ndsj/2024/indexch.htm (2025).
12. Liu, S. et al. Diversifying heat sources in China's urban district heating systems will reduce risk of carbon lock-in. *Nat. Energy* **9**, 1021–1031 (2024).
13. Hao, J., Yang, Y., Xu, C. & Du, X. A comprehensive review of planning, modeling, optimization, and control of distributed energy systems. *Carbon Neutrality* **1**, 28 (2022).
14. Mancarella, P. MES (multi-energy systems): an overview of concepts and evaluation models. *Energy* **65**, 1–17 (2014).
15. Cagnano, A., De Tuglie, E. & Mancarella, P. Microgrids: overview and guidelines for practical implementations and operation. *Appl. Energy* **258**, 114039 (2020).
16. Zhuo, Z. et al. Cost increase in the electricity supply to achieve carbon neutrality in China. *Nat. Commun.* **13**, 3172 (2022).
17. Tao, Y., Qiu, J., Lai, S. & Zhao, J. Integrated electricity and hydrogen energy sharing in coupled energy systems. *IEEE Trans. Smart Grid* **12**, 1149–1162 (2021).
18. Shahid, U. B., Bicer, Y., Ahzi, S. & Abdala, A. Thermodynamic assessment of an integrated renewable energy multigeneration system including ammonia as hydrogen carrier and phase change material energy storage. *Energy Convers. Manag.* **198**, 111809 (2019).
19. Lonis, F., Tola, V. & Cau, G. Assessment of integrated energy systems for the production and use of renewable methanol by water electrolysis and CO_2 hydrogenation. *Fuel* **285**, 119160 (2021).
20. Wang, J., Li, S., Zhang, G. & Yang, Y. Performance investigation of a solar-assisted hybrid combined cooling, heating and power system based on energy, exergy, exergo-economic and exergo-environmental analyses. *Energy Convers. Manag.* **196**, 227–241 (2019).
21. Jiang, J. et al. A review and perspective on industry high-temperature heat pumps. *Renew. Sustain. Energy Rev.* **161**, 112106 (2022).
22. Huang, Y., Wang, Y. & Liu, N. A two-stage energy management for heat-electricity integrated energy system considering dynamic pricing of Stackelberg game and operation strategy optimization. *Energy* **244**, 122576 (2022).
23. GeSI. *SMARTer2030* https://smarter2030.gesi.org/downloads/Full_report.pdf (2015).
24. Ding, C., Ke, J., Levine, M. & Zhou, N. Potential of artificial intelligence in reducing energy and carbon emissions of commercial buildings at scale. *Nat. Commun.* **15**, 5916 (2024).
25. Alabi, T. M. et al. A review on the integrated optimization techniques and machine learning approaches for modeling, prediction, and decision making on integrated energy systems. *Renew. Energy* **194**, 822–849 (2022).
26. Entezari, A., Aslani, A., Zahedi, R. & Noorollahi, Y. Artificial intelligence and machine learning in energy systems: a bibliographic perspective. *Energy Strategy Rev.* **45**, 101017 (2023).
27. Wang, R., Wen, X., Wang, X., Fu, Y. & Zhang, Y. Low carbon optimal operation of integrated energy system based on carbon capture technology, LCA carbon emissions and ladder-type carbon trading. *Appl. Energy* **311**, 118664 (2022).
28. Wang, Y., Qiu, D., Sun, M., Strbac, G. & Gao, Z. Secure energy management of multi-energy microgrid: a physical-informed safe reinforcement learning approach. *Appl. Energy* **335**, 120759 (2023).
29. Song, Y., Mu, H., Li, N., Wang, H. & Kong, X. Optimal scheduling of zero-carbon integrated energy system considering long- and short-term energy storages, demand response, and uncertainty. *J. Clean. Prod.* **435**, 140393 (2024).
30. Qu, Z., Xu, C., Yang, F., Ling, F. & Pirouzi, S. Market clearing price-based energy management of grid-connected renewable energy hubs including flexible sources according to thermal, hydrogen, and compressed air storage systems. *J. Energy Storage* **69**, 107981 (2023).
31. Majidi, H. et al. Overview of energy modeling requirements and tools for future smart energy systems. *Renew. Sustain. Energy Rev.* **212**, 115367 (2025).
32. Abugabbara, M., Javed, S. & Johansson, D. A simulation model for the design and analysis of district systems with simultaneous heating and cooling demands. *Energy* **261**, 125245 (2022).
33. Xu, L., Guo, Q., Wang, Z. & Sun, H. Modeling of time-delayed distributed cyber-physical power systems for small-signal stability analysis. *IEEE Trans. Smart Grid* **12**, 3425–3437 (2021).
34. Elaouzy, Y. & El Fadar, A. Investigation of building-integrated photovoltaic, photovoltaic thermal, ground source heat pump and green roof systems. *Energy Convers. Manag.* **283**, 116926 (2023).
35. Xie, H., Sun, X., Chen, C., Bie, Z. & Catalão, J. P. S. Resilience metrics for integrated power and natural gas systems. *IEEE Trans. Smart Grid* **13**, 2483–2486 (2022).
36. Steinegger, J., Wallner, S., Greiml, M. & Kienberger, T. A new quasi-dynamic load flow calculation for district heating networks. *Energy* **266**, 126410 (2023).
37. Zheng, C. et al. Energy flow optimization of integrated gas and power systems in continuous time and space. *IEEE Trans. Smart Grid* **12**, 2611–2624 (2021).
38. You, M., Wang, Q., Sun, H., Castro, I. & Jiang, J. Digital twins based day-ahead integrated energy system scheduling under load and renewable energy uncertainties. *Appl. Energy* **305**, 117899 (2022).
39. Chapaloglou, S. et al. Microgrid energy management strategies assessment through coupled thermal-electric considerations. *Energy Convers. Manag.* **228**, 113711 (2021).
40. Qureshi, Y., Ali, U. & Sher, F. Part load operation of natural gas fired power plant with CO_2 capture system for selective exhaust gas recirculation. *Appl. Therm. Eng.* **190**, 116808 (2021).
41. Li, G., Du, G., Liu, G. & Yan, J. Study on the dynamic characteristics, control strategies and load variation rates of the concentrated solar power plant. *Appl. Energy* **357**, 122538 (2024).
42. Lu, M. et al. Operational optimization of district heating system based on an integrated model in TRNSYS. *Energy Build.* **230**, 110538 (2021).
43. Wang, H. et al. A zoned group control of indoor temperature based on MPC for a space heating building. *Energy Convers. Manag.* **290**, 117196 (2023).
44. Alonso, M. J., Dols, W. S. & Mathisen, H. Using Co-simulation between EnergyPlus and CONTAM to evaluate recirculation-based, demand-controlled ventilation strategies in an office building. *Build. Environ.* **211**, 108737 (2022).
45. Wang, W. et al. Performance improvement of air-source heat pump heating system with variable water temperature difference. *Appl. Therm. Eng.* **210**, 118366 (2022).
46. Zimmerman, R. D., Murillo-Sánchez, C. E. & Thomas, R. J. MATPOWER: steady-state operations, planning, and analysis tools for power systems research and education. *IEEE Trans. Power Syst.* **26**, 12–19 (2011).
47. Song, Y., Chen, Y., Yu, Z., Huang, S. & Shen, C. CloudPSS: a high-performance power system simulator based on cloud computing. *Energy Rep.* **6**, 1611–1618 (2020).
48. Zhu, R. et al. GIScience can facilitate the development of solar cities for energy transition. *Adv. Appl. Energy* **10**, 100129 (2023).

49. Koirala, B. et al. Digitalization of urban multi-energy systems—advances in digital twin applications across life-cycle phases. *Adv. Appl. Energy* **16**, 100196 (2024).
50. Hilpert, S. et al. The Open Energy Modelling Framework (OEMOF)—a new approach to facilitate open science in energy system modelling. *Energy Strategy Rev.* **22**, 16–25 (2018).
51. Howells, M. et al. OSeMOSYS: the open source energy modeling system: an introduction to its ethos, structure and development. *Energy Policy* **39**, 5850–5870 (2011).
52. Lund, H. et al. EnergyPLAN—advanced analysis of smart energy systems. *Smart Energy* **1**, 100007 (2021).
53. Hunter, K., Sreepathi, S. & DeCarolis, J. F. Modeling for insight using tools for energy model optimization and analysis (TEMOA). *Energy Econ.* **40**, 339–349 (2013).
54. Berendes, S. et al. Evaluating the usability of open source frameworks in energy system modelling. *Renew. Sustain. Energy Rev.* **159**, 112174 (2022).
55. Groissböck, M. Are open source energy system optimization tools mature enough for serious use?. *Renew. Sustain. Energy Rev.* **102**, 234–248 (2019).
56. Geidl, M. et al. Energy hubs for the future. *IEEE Power Energy Mag.* **5**, 24–30 (2007).
57. Mansouri, S. A. et al. A Multi-objective dynamic framework for design of energy hub by considering energy storage system, power-to-gas technology and integrated demand response program. *J. Energy Storage* **50**, 104206 (2022).
58. Akbari, E., Shabestari, S. F. M., Pirouzi, S. & Jadidoleslam, M. Network flexibility regulation by renewable energy hubs using flexibility pricing-based energy management. *Renew. Energy* **206**, 295–308 (2023).
59. Wang, L. et al. Optimal operation analysis of integrated community energy system considering the uncertainty of demand response. *IEEE Trans. Power Syst.* **36**, 3681–3691 (2021).
60. Oskouei, M. Z., Mohammadi-Ivatloo, B., Abapour, M., Shafiee, M. & Anvari-Moghaddam, A. Techno-economic and environmental assessment of the coordinated operation of regional grid-connected energy hubs considering high penetration of wind power. *J. Clean. Prod.* **280**, 124275 (2021).
61. Nozari, M. H., Yaghoubi, M., Jafarpur, K. & Mansoori, G. A. Development of dynamic energy storage hub concept: a comprehensive literature review of multi storage systems. *J. Energy Storage* **48**, 103972 (2022).
62. Mohammadi, M., Noorollahi, Y., Mohammadi-Ivatloo, B. & Yousefi, H. Energy hub: from a model to a concept—a review. *Renew. Sustain. Energy Rev.* **80**, 1512–1527 (2017).
63. Pazmiño-Arias, A., Briceño-León, M., Clairand, J.-M., Serrano-Guerrero, X. & Escrivá-Escrivá, G. Optimal scheduling of a dairy industry based on energy hub considering renewable energy and ice storage. *J. Clean. Prod.* **429**, 139580 (2023).
64. Zhang, H., Sun, W., Li, W. & Ma, G. A carbon flow tracing and carbon accounting method for exploring CO2 emissions of the iron and steel industry: an integrated material–energy–carbon hub. *Appl. Energy* **309**, 118485 (2022).
65. Lasemi, M. A., Arabkoohsar, A., Hajizadeh, A. & Mohammadi-Ivatloo, B. A comprehensive review on optimization challenges of smart energy hubs under uncertainty factors. *Renew. Sustain. Energy Rev.* **160**, 112320 (2022).
66. Jin, Y. et al. A prospective review on the design and operation of integrated energy system: the spotlight cast on dynamic characteristics. *Appl. Thermal Eng.* **253**, 123751 (2024).
67. Yang, J., Zhang, N., Botterud, A. & Kang, C. On an equivalent representation of the dynamics in district heating networks for combined electricity-heat operation. *IEEE Trans. Power Syst.* **35**, 560–570 (2019).
68. Yang, J., Zhang, N., Botterud, A. & Kang, C. Situation awareness of electricity-gas coupled systems with a multi-port equivalent gas network model. *Appl. Energy* **258**, 114029 (2020).
69. Chen, B., Wu, W., Guo, Q. & Sun, H. An efficient optimal energy flow model for integrated energy systems based on energy circuit modeling in the frequency domain. *Appl. Energy* **326**, 119923 (2022).
70. Zhang, S. et al. Unified modeling of integrated energy networks in time domain and its applications (I): two-port models in time domain. *Proc. CSEE* **41**, 6509–6521 (2021).
71. Zhang, S. et al. Unified modeling of integrated energy networks in time domain and its applications (II): augmented model in complex frequency domain and comparison between time/frequency domain models. *Proc. CSEE* **42**, 5798–5810 (2021).
72. Chen, Q., Fu, R. & Xu, Y. Electrical circuit analogy for heat transfer analysis and optimization in heat exchanger networks. *Appl. Energy* **139**, 81–92 (2015).
73. Chen, Q., Meng, N., He, K.-L., Ma, H. & Gou, X. Multi-time scale operation optimization of integrated power and thermal system considering load disturbance. *Energy* **302**, 131935 (2024).
74. Yao, Z. et al. Machine learning for a sustainable energy future. *Nat. Rev. Mater.* **8**, 202–215 (2023).
75. Lin, X. et al. Component modeling and updating method of integrated energy systems based on knowledge distillation. *Energy AI* **16**, 100350 (2024).
76. Li, W., Chen, L., Zhao, J. & Wang, W. Mechanism and data-driven dual estimation of coupling hydraulic-thermal states for steam heating networks considering multi-time-scale characteristics. *IEEE Trans. Cybern.* **54**, 4567–4580 (2023).
77. Jiang, M., Speetjens, M., Rindt, C. & Smeulders, D. A data-based reduced-order model for dynamic simulation and control of district-heating networks. *Appl. Energy* **340**, 121038 (2023).
78. Mehdi, S. & Tiwary, P. Thermodynamics-inspired explanations of artificial intelligence. *Nat. Commun.* **15**, 7859 (2024).
79. Li, J. & Guo, L. The logic and architecture of future data systems. *Engineering* **47**, 14–15 (2025).
80. Sun, Y., Sengupta, U. & Juniper, M. Physics-informed deep learning for simultaneous surrogate modeling and PDE-constrained optimization of an airfoil geometry. *Comput. Methods Appl. Mech. Eng.* **411**, 116042 (2023).
81. Lee, J. et al. Extended multiphysics-informed neural network for conjugate heat transfer problems. *Int. J. Heat Mass Transf.* **246**, 127098 (2025).
82. Wu, H. & Xu, Z. Multi-energy flow calculation in integrated energy system via topological graph attention convolutional network with transfer learning. *Energy* **303**, 132018 (2024).
83. Ngo, Q.-H., Nguyen, B. L. H., Vu, T. V., Zhang, J. & Ngo, T. Physics-informed graphical neural network for power system state estimation. *Appl. Energy* **358**, 122602 (2024).
84. Raissi, M., Perdikaris, P. & Karniadakis, G. E. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *J. Comput. Phys.* **378**, 686–707 (2019).
85. Huang, B. & Wang, J. Applications of physics-informed neural networks in power systems— a review. *IEEE Trans. Power Syst.* **38**, 572–588 (2023).
86. Xiao, T. & You, F. Physically consistent deep learning-based day-ahead energy dispatching and thermal comfort control for grid-interactive communities. *Appl. Energy* **353**, 122133 (2024).
87. Gokhale, G., Claessens, B. & Develder, C. Physics informed neural networks for control oriented thermal modeling of buildings. *Appl. Energy* **314**, 118852 (2022).
88. Ishitsuka, K. & Lin, W. Physics-informed neural network for inverse modeling of natural-state geothermal systems. *Appl. Energy* **337**, 120855 (2023).

89. Zhuang, W., Fan, J., Xia, M. & Zhu, K. A multi-scale spatial–temporal graph neural network-based method of multienergy load forecasting in integrated energy system. *IEEE Trans. Smart Grid* **15**, 2652–2666 (2024).
90. Wu, Z. et al. A comprehensive survey on graph neural networks. *IEEE Trans. Neural Netw. Learn. Syst.* **32**, 4–24 (2021).
91. Yang, Z. et al. Physics-constrained graph modeling for building thermal dynamics. *Energy AI* **16**, 100346 (2024).
92. Yang, M. et al. Topology-transferable physics-guided graph neural network for real-time optimal power flow. *IEEE Trans. Ind. Inform.* **20**, 10857–10872 (2024).
93. Boussaid, T., Rousset, F., Scuturici, V.-M. & Clausse, M. Enabling fast prediction of district heating networks transients via a physics-guided graph neural network. *Appl. Energy* **370**, 123634 (2024).
94. Dan, Z., Song, A., Zheng, Y., Zhang, X. & Zhou, Y. City information models for optimal EV charging and energy-resilient renaissance. *Nexus* **2**, 100056 (2025).
95. Madler, J., Harding, S. & Weibelzahl, M. A multi-agent model of urban microgrids: assessing the effects of energy-market shocks using real-world data. *Appl. Energy* **343**, 121180 (2023).
96. Ball, P. D., Evans, S., Levers, A. & Ellison, D. Zero carbon manufacturing facility—towards integrating material, energy, and waste process flows. *Proc. Inst. Mech. Eng. Part B: J. Eng. Manuf.* **223**, 1085–1096 (2009).
97. Calise, F. et al. Optimal operating strategies of combined cooling, heating and power systems: a case study for an engine manufacturing facility. *Energy Convers. Manag.* **149**, 1066–1084 (2017).
98. Eardley, S., Choi, J.-K., Hong, T. & An, J. Decarbonization potential of regional combined heat and power development. *Renew. Sustain. Energy Rev.* **189**, 114038 (2024).
99. Ranaboldo, M. et al. A comprehensive overview of industrial demand response status in Europe. *Renew. Sustain. Energy Rev.* **203**, 114797 (2024).
100. West, K., van Harmelen, T., Koning, V., Kramer, G. J. & Faaij, A. Detailed modelling of basic industry and material flows in a national energy system optimization model. *Resour. Conserv. Recycl.* **206**, 107617 (2024).
101. Wesseling, J. H. et al. The transition of energy intensive processing industries towards deep decarbonization: characteristics and implications for future research. *Renew. Sustain. Energy Rev.* **79**, 1303–1313 (2017).
102. Golmohamadi, H. Demand-side management in industrial sector: a review of heavy industries. *Renew. Sustain. Energy Rev.* **156**, 111963 (2022).
103. Mossie, A. T., Khatiwada, D., Palm, B. & Bekele, G. Energy demand flexibility potential in cement industries: how does it contribute to energy supply security and environmental sustainability?. *Appl. Energy* **377**, 124608 (2025).
104. Wang, J., Wang, Q. & Sun, W. Quantifying flexibility provisions of the ladle furnace refining process as cuttable loads in the iron and steel industry. *Appl. Energy* **342**, 121178 (2023).
105. Zhang, Y. et al. Multi-time scale customer directrix load-based demand response under renewable energy and customer uncertainties. *Appl. Energy* **383**, 125334 (2025).
106. Tang, Q. Q. et al. Design optimization of industrial energy systems with energy consumption relaxation models for coupling process units and utility streams. *J. Clean. Prod.* **344**, 131072 (2022).
107. Leherbauer, D., Schulz, J., Egyed, A. & Hehenberger, P. Demand-side management in less energy-intensive industries: a systematic mapping study. *Renew. Sustain. Energy Rev.* **212**, 115315 (2025).
108. Mu, J. et al. Complex coupling representation in low-dimensional space for control-oriented energy-consuming industries modeling. *Appl. Energy* **383**, 125263 (2025).
109. Xu, T., Long, J., Zhao, L. & Du, W. Material and energy coupling systems optimization for large-scale industrial refinery with sustainable energy penetration under multiple uncertainties using two-stage stochastic programming. *Appl. Energy* **371**, 123525 (2024).
110. Gutowski, T. G. et al. Thermodynamic analysis of resources used in manufacturing processes. *Environ. Sci. Technol.* **43**, 1584–1590 (2009).
111. Menghi, R., Papetti, A., Germani, M. & Marconi, M. Energy efficiency of manufacturing systems: a review of energy assessment methods and tools. *J. Clean. Prod.* **240**, 118276 (2019).
112. Yuksek, Y. A., Haddad, Y., Cox, R. & Salonitis, K. A unit product energy mapping framework for operation management in manufacturing industries. *Procedia CIRP* **122**, 873–878 (2024).
113. Gonçalves, S. N., Albuquerque, D. M. & Pereira, J. C. Modelling and energy efficiency analysis of the microwave continuous processing of limestone. *J. Clean. Prod.* **467**, 142912 (2024).
114. Kim, S. & Joo, Y. Energy-based key performance indicator for energy-intensive manufacturing processes: application to steel casting. *Energy* **317**, 134543 (2025).
115. Hu, H. et al. Carbon-material-energy flows nexus analysis of electric arc furnace steelmaking processes in China. *J. Clean. Prod.* **486**, 144377 (2025).
116. Karimi-Zare, A., Shakouri, H., Kazemi, A. & Kim, E.-S. Aggregate production planning and energy supply management in steel industry with an onsite energy generation system: a multi-objective robust optimization model. *Int. J. Prod. Econ.* **269**, 109149 (2024).
117. Zhang, T., Shen, F., Peng, X., Li, Z. & Zhong, W. Carbon-efficient production planning for long-chain integrated refinery-petrochemical processes: a material-energy-carbon optimization perspective. *J. Clean. Prod.* **426**, 138916 (2023).
118. Zwickl-Bernhard, S. Integrating material recycling and remanufacturing in energy system optimization modeling: a review and showcase. *Adv. Appl. Energy* 100198 (2024).
119. Ding, Y. M., Hong, S. H. & Li, X. H. A demand response energy management scheme for industrial facilities in smart grid. *IEEE Trans. Ind. Inform.* **10**, 2257–2269 (2014).
120. Huang, B., Zhou, M., Lu, X. S. & Abusorrah, A. Scheduling of resource allocation systems with timed Petri nets: a survey. *ACM Comput. Surv.* **55**, 1–27 (2023).
121. Lu, X., Zhou, M., Ammari, A. C. & Ji, J. Hybrid Petri nets for modeling and analysis of microgrid systems. *IEEE/CAA J. Autom. Sin.* **3**, 349–356 (2016).
122. Li, C. et al. Energy hub-based optimal planning for integrated energy systems considering part-load characteristics and synergistic effect of equipment. *Glob. Energy Interconnect.* **4**, 169–183 (2021).
123. Li, C. et al. Energy hub-based optimal planning framework for user-level integrated energy systems: considering synergistic effects under multiple uncertainties. *Appl. Energy* **307**, 118099 (2022).
124. Bao, M. et al. An efficient framework for exploiting operational flexibility of load energy hubs in risk management of integrated electricity-gas systems. *Appl. Energy* **338**, 120765 (2023).
125. Hui, H., Bao, M., Ding, Y. & Song, Y. Exploring the integrated flexible region of distributed multi-energy systems with process industry. *Appl. Energy* **311**, 118590 (2022).
126. Ding, Y. & Wei, X. Bi-level optimization model for regional energy system planning under demand response scenarios. *J. Clean. Prod.* **323**, 129009 (2021).
127. He, J., Wu, Y., Yong, X., Tan, Q. & Liu, F. Bi-level optimization of a near-zero-emission integrated energy system considering electricity-hydrogen-gas nexus: a two-stage framework aiming at economic and environmental benefits. *Energy Convers. Manag.* **274**, 116434 (2022).

128. Zhang, Z. et al. Equalizing multi-temporal scale adequacy for low carbon power systems by co-planning short-term and seasonal energy storage. *J. Energy Storage* **84**, 111518 (2024).
129. Zhou, S. et al. Risk-averse bi-level planning model for maximizing renewable energy hosting capacity via empowering seasonal hydrogen storage. *Appl. Energy* **361**, 122853 (2024).
130. Zhang, L., Sun, B. & Li, F. Triple-layer joint optimization of capacity and operation for integrated energy systems by coordination on multiple timescales. *Energy* **302**, 131775 (2024).
131. Lin, X., Zhang, N., Zhong, W., Kong, F. & Cong, F. Regional integrated energy system long-term planning optimization based on multi-energy complementarity quantification. *J. Build. Eng.* **68**, 106046 (2023).
132. Langiu, M. et al. COMANDO: a next-generation open-source framework for energy systems optimization. *Comput. Chem. Eng.* **152**, 107366 (2021).
133. Qin, C., Yan, Q. & He, G. Integrated energy systems planning with electricity, heat and gas using particle swarm optimization. *Energy* **188**, 116044 (2019).
134. Li, Q. et al. Hierarchical optimal scheduling method for regional integrated energy systems considering electricity-hydrogen shared energy. *Appl. Energy* **349**, 121670 (2023).
135. Medghalchi, Z. & Taylan, O. A novel hybrid optimization framework for sizing renewable energy systems integrated with energy storage systems with solar photovoltaics, wind, battery and electrolyzer-fuel cell. *Energy Convers. Manag.* **294**, 117594 (2023).
136. Hu, H., Sun, X., Zeng, B., Gong, D. & Zhang, Y. Enhanced evolutionary multi-objective optimization-based dispatch of coal mine integrated energy system with flexible load. *Appl. Energy* **307**, 118130 (2022).
137. Yan, R. et al. Flexibility improvement and stochastic multi-scenario hybrid optimization for an integrated energy system with high-proportion renewable energy. *Energy* **263**, 125779 (2023).
138. Fan, W. et al. Two-stage distributionally robust optimization model of integrated energy system group considering energy sharing and carbon transfer. *Appl. Energy* **331**, 120426 (2023).
139. Yang, W., Huang, Y., Ye, T., Pan, F. & Zhao, D. Hybrid continuous-discrete time control strategy to optimize thermal network dynamic storage and heat-electricity integrated energy system dynamic operation. *Energy Convers. Manag.* **301**, 118035 (2024).
140. Shi, M. et al. Distributed energy scheduling for integrated energy system clusters with peer-to-peer energy transaction. *IEEE Trans. Smart Grid* **14**, 142–156 (2023).
141. Li, P., Wang, Z., Wang, J., Guo, T. & Yin, Y. A multi-time-space scale optimal operation strategy for a distributed integrated energy system. *Appl. Energy* **289**, 116698 (2021).
142. Wang, L., Lin, J., Dong, H., Wang, Y. & Zeng, M. Demand response comprehensive incentive mechanism-based multi-time scale optimization scheduling for park integrated energy system. *Energy* **270**, 126893 (2023).
143. Wang, Z. et al. Multi-time scale scheduling optimization of integrated energy systems considering seasonal hydrogen utilization and multiple demand responses. *Int. J. Hydrog. Energy* **67**, 728–749 (2024).
144. Shirizadeh, B. & Quirion, P. Do multi-sector energy system optimization models need hourly temporal resolution? A case study with an investment and dispatch model applied to France. *Appl. Energy* **305**, 117951 (2022).
145. Guo, H. et al. Dynamic characteristics and control of supercritical compressed air energy storage systems. *Appl. Energy* **283**, 116294 (2021).
146. Zhang, K., Liu, M., Zhao, Y., Yan, H. & Yan, J. Design and performance evaluation of a new thermal energy storage system integrated within a coal-fired power plant. *J. Energy Storage* **50**, 104335 (2022).
147. Ji, W. et al. Applications of flywheel energy storage system on load frequency regulation combined with various power generations: a review. *Renew. Energy* **223**, 119975 (2024).
148. Li, H. et al. Energy and entropy generation analysis in a supercapacitor for different operating conditions. *Energy* **260**, 124932 (2022).
149. Shen, W., Zeng, B. & Zeng, M. Multi-timescale rolling optimization dispatch method for integrated energy system with hybrid energy storage system. *Energy* **283**, 129006 (2023).
150. Wang, Y. et al. Research on capacity planning and optimization of regional integrated energy system based on hybrid energy storage system. *Appl. Therm. Eng.* **180**, 115834 (2020).
151. Zhu, Y. et al. Optimal dispatch of a novel integrated energy system combined with multi-output organic Rankine cycle and hybrid energy storage. *Appl. Energy* **343**, 121113 (2023).
152. Liu, Z. et al. A novel distributed energy system combining hybrid energy storage and a multi-objective optimization method for nearly zero-energy communities and buildings. *Energy* **239**, 122577 (2022).
153. Hu, J., Wang, Y. & Dong, L. Low carbon-oriented planning of shared energy storage station for multiple integrated energy systems considering energy-carbon flow and carbon emission reduction. *Energy* **290**, 130139 (2024).
154. Wang, S., Zhai, J., Hui, H., Ding, Y. & Song, Y. Operational reliability of integrated energy systems considering gas flow dynamics and demand-side flexibilities. *IEEE Trans. Ind. Inform.* **20**, 1360–1373 (2024).
155. Wang, Y., Dong, H., Ma, K., Wang, H. & Zhang, J. Multi frequency stability optimization of integrated energy systems considering virtual energy storage characteristics of heating networks. *Appl. Therm. Eng.* **257**, 124254 (2024).
156. Hu, Z. et al. Techno-economic modeling and safe operational optimization of multi-network constrained integrated community energy systems. *Adv. Appl. Energy* **15**, 100183 (2024).
157. Gu, W., Wang, J., Lu, S., Luo, Z. & Wu, C. Optimal operation for integrated energy system considering thermal inertia of district heating network and buildings. *Appl. Energy* **199**, 234–246 (2017).
158. Pan, Z., Guo, Q. & Sun, H. Feasible region method based integrated heat and electricity dispatch considering building thermal inertia. *Appl. Energy* **192**, 395–407 (2017).
159. Liu, X. et al. Multi-time scale optimal scheduling of integrated electricity and district heating systems considering thermal comfort of users: an enhanced-interval optimization method. *Energy* **254**, 124311 (2022).
160. Zhong, W., Zhang, S., Lin, X., Wang, J. & Zhou, Y. Quantitative study on steam system thermal inertia based on a dynamic hydraulic analytical model. *J. Clean. Prod.* **403**, 136864 (2023).
161. Zhuang, W., Zhou, S., Chen, J. & Gu, W. Operation optimization of electricity-steam coupled industrial energy system considering steam accumulator. *Energy* **289**, 129903 (2024).
162. Lund, P. D., Lindgren, J., Mikkola, J. & Salpakari, J. Review of energy system flexibility measures to enable high levels of variable renewable electricity. *Renew. Sustain. energy Rev.* **45**, 785–807 (2015).
163. Vandermeulen, A., van der Heijde, B. & Helsen, L. Controlling district heating and cooling networks to unlock flexibility: a review. *Energy* **151**, 103–115 (2018).
164. Tang, H., Wang, S. & Li, H. Flexibility categorization, sources, capabilities and technologies for energy-flexible and grid-responsive buildings: state-of-the-art and future perspective. *Energy* **219**, 119598 (2021).
165. Liu, X., Zhang, L., Jin, K., Xue, X. & Zhou, H. Integration model and performance analysis of coupled thermal energy storage and ejector flexibility retrofit for 600 MW thermal power units. *J. Clean. Prod.* **428**, 139337 (2023).

166. Wang, H. et al. Heat-power decoupling and energy saving of the CHP unit with heat pump based waste heat recovery system. *Energy* **250**, 123846 (2022).
167. Ma, T. et al. Design and performance analysis of deep peak shaving scheme for thermal power units based on high-temperature molten salt heat storage system. *Energy* **288**, 129557 (2024).
168. Zhang, Y., Johansson, P. & Kalagasidis, A. S. Feasibilities of utilizing thermal inertia of district heating networks to improve system flexibility. *Appl. Therm. Eng.* **213**, 118813 (2022).
169. Vivian, J., Quaggiotto, D. & Zarrella, A. Increasing the energy flexibility of existing district heating networks through flow rate variations. *Appl. Energy* **275**, 115411 (2020).
170. Hammer, A., Sejkora, C. & Kienberger, T. Increasing district heating networks efficiency by means of temperature-flexible operation. *Sustain. Energy Grids Netw.* **16**, 393–404 (2018).
171. Zhong, W., Chen, J., Zhou, Y., Li, Z. & Lin, X. Network flexibility study of urban centralized heating system: concept, modeling and evaluation. *Energy* **177**, 334–346 (2019).
172. Srithapon, C. & Månsson, D. Predictive control and coordination for energy community flexibility with electric vehicles, heat pumps and thermal energy storage. *Appl. Energy* **347**, 121500 (2023).
173. Nozarian, M., Hajizadeh, A. & Fereidunian, A. A methodological review on management of building cluster meso energy hubs in accordance with the agile framework: exploring flexibility upward the building-cluster-city hierarchy. *Sustain. Energy Technol. Assess.* **67**, 103834 (2024).
174. Zhou, Y. & Cao, S. Quantification of energy flexibility of residential net-zero-energy buildings involved with dynamic operations of hybrid energy storages and diversified energy conversion strategies. *Sustain. Energy Grids Netw.* **21**, 100304 (2020).
175. Guo, Y. et al. Key district heating technologies for building energy flexibility: a review. *Renew. Sustain. Energy Rev.* **189**, 114017 (2024).
176. Pastore, L. M. Combining Power-to-Heat and Power-to-Vehicle strategies to provide system flexibility in smart urban energy districts. *Sustain. Cities Soc.* **94**, 104548 (2023).
177. Zhou, Y. Low-carbon transition in smart city with sustainable airport energy ecosystems and hydrogen-based renewable-grid-storage-flexibility. *Energy Rev.* **1**, 100001 (2022).
178. Xu, X., Sun, W., Abeysekera, M. & Qadrdan, M. Quantifying the flexibility from industrial steam systems for supporting the power grid. *IEEE Trans. Power Syst.* **36**, 313–322 (2021).
179. Tie, Y. et al. Integrated flexibility characterization and measurement of distributed multi-energy systems considering temporal coupling constraints. *Energy* **283**, 128684 (2023).
180. Liu, J., Ma, L. & Wang, Q. Energy management method of integrated energy system based on collaborative optimization of distributed flexible resources. *Energy* **264**, 125981 (2023).
181. Karimi, H. & Jadid, S. Multi-layer energy management of smart integrated-energy microgrid systems considering generation and demand-side flexibility. *Appl. Energy* **339**, 120984 (2023).
182. Gao, X., Knueven, B., Siirola, J. D., Miller, D. C. & Dowling, A. W. Multiscale simulation of integrated energy system and electricity market interactions. *Appl. Energy* **316**, 119017 (2022).
183. Ma, H. et al. A compact model to coordinate flexibility and efficiency for decomposed scheduling of integrated energy system. *Appl. Energy* **285**, 116474 (2021).
184. Du, Y., Yi, Q., Li, C. & Liao, L. Life cycle oriented low-carbon operation models of machinery manufacturing industry. *J. Clean. Prod.* **91**, 145–157 (2015).
185. Shang, L. et al. Pathways for supply security and carbon-neutral transition in the oil products industry: a comprehensive technology portfolio evaluation. *J. Clean. Prod.* **499**, 145241 (2025).
186. Wu, W. et al. Industrial Park low-carbon energy system planning framework: heat pump based energy conjugation between industry and buildings. *Appl. Energy* **369**, 123594 (2024).
187. Shen, A. & Zhang, J. Technologies for CO_2 emission reduction and low-carbon development in primary aluminum industry in China: a review. *Renew. Sustain. Energy Rev.* **189**, 113965 (2024).
188. Chen, J. et al. An integrated design method for CO_2 emission reduction based on industrial metabolism: hybrid carbon-hydrogen metallurgy manufacturing process (HCHMP). *J. Clean. Prod.* **486**, 144595 (2025).
189. Ledari, M. B. et al. Advanced utilization of urban sewage sludge for low-carbon hydrogen and power production in steel manufacturing. *Energy Convers. Manag.* **315**, 118657 (2024).
190. Wang, P., Xu, Q., Wang, F. & Xu, M. Study on the coupling of the iron and steel industry with renewable energy for low-carbon production: a case study of matching steel plants with photovoltaic power plants in China. *Energy* **320**, 135381 (2025).
191. Jiang, M. et al. Tracing fossil-based plastics, chemicals and fertilizers production in China. *Nat. Commun.* **15**, 3854 (2024).
192. Jost, D., Kanzurova, S., Nilges, B., Reinert, C. & von der Assen, N. Life cycle assessment of measures towards a low-carbon flat glass production. *J. Clean. Prod.* **501**, 145294 (2025).
193. Meys, R. et al. Achieving net-zero greenhouse gas emission plastics by a circular carbon economy. *Science* **374**, 71–76 (2021).
194. Zanon-Zotin, M. et al. Unaddressed non-energy use in the chemical industry can undermine fossil fuels phase-out. *Nat. Commun.* **15**, 8050 (2024).
195. Satymov, R., Bogdanov, D., Galimova, T. & Breyer, C. Energy and industry transition to carbon-neutrality in Nordic conditions via local renewable sources, electrification, sector coupling and Power-to-X. *Energy* **319**, 134888 (2025).
196. Verpoort, P. C., Gast, L., Hofmann, A. & Ueckerdt, F. Impact of global heterogeneity of renewable energy supply on heavy industrial production and green value chains. *Nat. Energy* **9**, 491–503 (2024).
197. Liu, C. et al. Emergy-based evaluation and improvement for sustainable manufacturing systems considering resource efficiency and environment performance. *Energy Convers. Manag.* **177**, 176–189 (2018).
198. Yang, M. & Liu, Y. Research on multi-energy collaborative operation optimization of integrated energy system considering carbon trading and demand response. *Energy* **283**, 129117 (2023).
199. Mengesha, I. & Roy, D. Carbon pricing drives critical transition to green growth. *Nat. Commun.* **16**, 1321 (2025).
200. Fisco-Compte, P. et al. Sustainable optimal production scheduling methodology under implicit demand response and carbon policy: a manufacturing industry case study. *Appl. Energy* **384**, 125403 (2025).
201. Li, H. et al. Process analysis-based industrial production modelling with uncertainty: a linear fractional programming for joint optimization of total carbon emissions and emission intensity. *Appl. Energy* **382**, 125204 (2025).
202. Ma, S. et al. Low-carbon economic operation of energy hub integrated with linearization model and nodal energy-carbon price. *Energy* **294**, 130754 (2024).
203. Wu, H. et al. Low-carbon economic dispatch of iron and steel industry empowered by wind-hydrogen energy: Modeling and stochastic programming. *Appl. Energy* **387**, 125599 (2025).
204. Zhang, N. et al. Coordinated scheduling and carbon flow tracing of discrete manufacturing processes with integrated energy-material-carbon flows model. *Appl. Energy* **398**, 126435 (2025).
205. Fang, K., Uhan, N., Zhao, F. & Sutherland, J. W. A new approach to scheduling in manufacturing for power consumption and carbon footprint reduction. *J. Manuf. Syst.* **30**, 234–240 (2011).
206. Piroozfard, H., Wong, K. Y. & Wong, W. P. Minimizing total carbon footprint and total late work criterion in flexible job shop scheduling by using an improved multi-objective genetic algorithm. *Resour. Conserv. Recycl.* **128**, 267–283 (2018).

207. Wang, W. & Zhao, J. An enhanced memetic algorithm for energy-efficient and low-carbon flexible job shop scheduling problem considering machine restart. *J. Manuf. Syst.* **80**, 457–478 (2025).
208. Du, J., Wang, J., Liang, J. & Liang, R. Research on the impact of smart logistics on the manufacturing industry chain resilience. *Sci. Rep.* **15**, 9052 (2025).
209. Li, W., Li, H., Han, Y. & Wang, Y. An improved multi-objective evolutionary algorithm for the low-carbon flexible job shop scheduling with automated guided vehicles. *Appl. Soft Comput.* **175**, 113048 (2025).
210. Huang, S., Lu, H., Chen, M. & Zhao, W. Integrated energy system scheduling considering the correlation of uncertainties. *Energy* **283**, 129011 (2023).
211. Li, Y., Wang, B., Yang, Z., Li, J. & Chen, C. Hierarchical stochastic scheduling of multi-community integrated energy systems in uncertain environments via Stackelberg game. *Appl. Energy* **308**, 118392 (2022).
212. Zhang, Y. et al. Encoding frequency constraints in preventive unit commitment using deep learning with region-of-Interest active sampling. *IEEE Trans. Power Syst.* **37**, 1942–1955 (2022).
213. Ramesh, A. V. & Li, X. Spatio-temporal deep learning-assisted reduced security-constrained unit commitment. *IEEE Trans. Power Syst.* **39**, 4735–4746 (2024).
214. Kaelbling, L. P., Littman, M. L. & Moore, A. W. Reinforcement learning: a survey. *J. Artif. Intell. Res.* **4**, 237–285 (1996).
215. Henderson, P., Islam, R., Bachman, P., Pineau, J., Precup, D. & Meger, D. Deep reinforcement learning that matters. *Proc. AAAI Conf. Artif. Intell*. **32**, 3207–3214 (2018).
216. Lissa, P. et al. Deep reinforcement learning for home energy management system control. *Energy AI* **3**, 100043 (2021).
217. Chen, T. et al. Peer-to-peer energy trading and energy conversion in interconnected multi-energy microgrids using multi-agent deep reinforcement learning. *IEEE Trans. Smart Grid* **13**, 715–727 (2022).
218. Alqahtani, M. & Hu, M. Dynamic energy scheduling and routing of multiple electric vehicles using deep reinforcement learning. *Energy* **244**, 122626 (2022).
219. Perera, A. T. D. & Kamalaruban, P. Applications of reinforcement learning in energy systems. *Renew. Sustain. Energy Rev.* **137**, 110618 (2021).
220. Yang, T., Zhao, L., Li, W. & Zomaya, A. Y. Dynamic energy dispatch strategy for integrated energy system based on improved deep reinforcement learning. *Energy* **235**, 121377 (2021).
221. Zhang, B. et al. Soft actor-critic-based multi-objective optimized energy conversion and management strategy for integrated energy systems with renewable energy. *Energy Convers. Manag.* **243**, 114381 (2021).
222. Hanifi, S., Cammarono, A. & Zare-Behtash, H. Advanced hyperparameter optimization of deep learning models for wind power prediction. *Renew. Energy* **221**, 119700 (2024).
223. Schweidtmann, A. M. et al. Machine learning meets continuous flow chemistry: Automated optimization towards the Pareto front of multiple objectives. *Chem. Eng. J.* **352**, 277–282 (2018).
224. Tang, K. & Yao, X. Learn to optimize—a brief overview. *Natl. Sci. Rev.* **11**, nwae132 (2024).
225. Min, B. et al. Recent advances in natural language processing via large pre-trained language models: A survey. *ACM Comput. Surv.* **56**, 1–40 (2023).
226. Qu, L., Wu, S., Fei, H., Nie, L. & Chua, T.-S. LayoutLLM-T2I: Eliciting layout guidance from LLM for text-to-image generation. In *Proceedings of the 31st ACM International Conference on Multimedia*. 643–654 (ACM, 2023).
227. Huang, B., Wang, X., Chen, H., Song, Z. & Zhu, W. Vtimellm: Empower LLM to grasp video moments. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit*. 14271–14280 (IEEE, 2024).
228. Yan, Z. & Xu, Y. Real-time optimal power flow with linguistic stipulations: integrating GPT-agent and deep reinforcement learning. *IEEE Trans. Power Syst.* **39**, 4747–4750 (2024).
229. Zhu, G. et al. CMLLM: a novel cross-modal large language model for wind power forecasting. *Energy Convers. Manag.* **330**, 119673 (2025).
230. Xie, R. et al. Large language model-aided edge learning in distribution system state estimation. *IEEE Internet Things J.* 1–1 https://doi.org/10.1109/JIOT.2024.3483441 (2024).
231. Majumder, S. et al. Exploring the capabilities and limitations of large language models in the electric energy sector. *Joule* **8**, 1544–1549 (2024).
232. Liang, M., Hu, Y., Weng, H., Xi, J. & Yin, B. EnergyGPT: fine-tuning large language model for multi-energy load forecasting. *Renew. Energy* **251**, 123313 (2025).
233. Jiang, G., Ma, Z., Zhang, L. & Chen, J. EPlus-LLM: a large language model-based computing platform for automated building energy modeling. *Appl. Energy* **367**, 123431 (2024).
234. Li, Y. et al. A large language model-based building operation and maintenance information query. *Energy Build.* **334**, 115515 (2025).
235. Zhang, J., Zhang, C., Lu, J. & Zhao, Y. Domain-specific large language models for fault diagnosis of heating, ventilation, and air conditioning systems by labeled-data-supervised fine-tuning. *Appl. Energy* **377**, 124378 (2025).
236. Lai, Z., Wu, T., Fei, X. & Ling, Q. BERT4ST:: fine-tuning pre-trained large language model for wind power forecasting. *Energy Convers. Manag.* **307**, 118331 (2024).
237. Li, S., Huang, Z., Li, Y., Deng, S. & Cao, X. E. Methodology for predicting material performance by context-based modeling: a case study on solid amine CO_2 adsorbents. *Energy AI* **20**, 100477 (2025).
238. Zhang, L. & Chen, Z. Large language model-based interpretable machine learning control in building energy systems. *Energy Build.* **313**, 114278 (2024).
239. Zhang, X., Glaws, A., Cortiella, A., Emami, P. & King, R. N. Deep generative models in energy system applications: review, challenges, and future directions. *Appl. Energy* **380**, 125059 (2025).
240. Gao, C. et al. Large language models empowered agent-based modeling and simulation: a survey and perspectives. *Humanit. Soc. Sci. Commun.* **11**, 1–24 (2024).
241. Kar, I., Ralte, Z., Shivakumara, M., Roy, R. & Kumari, A. Agents are all you need: elevating trading dynamics with advanced generative AI-driven conversational LLM agents and tools. In *2024 IEEE 9th International Conference for Convergence in Technology (I2CT)*. 1–6 (IEEE, 2024).
242. Pereira, H., Gomes, L. & Vale, Z. Peer-to-peer energy trading optimization in energy communities using multi-agent deep reinforcement learning. *Energy Inform.* **5**, 44 (2022).
243. Yang, X., Lin, C., Liu, H. & Wu, W. R. L. 2: Reinforce large language model to assist safe reinforcement learning for energy management of active distribution networks. *IEEE Trans. Smart Grid* **16**, 3419–3431 (2025).
244. Xu, L., Lin, N., Poor, H. V., Xi, D. & Perera, A. T. D. Quantifying cascading power outages during climate extremes considering renewable energy integration. *Nat. Commun.* **16**, 2582 (2025).
245. Jiang, P., Sonne, C., Li, W., You, F. & You, S. Preventing the immense increase in the life-cycle energy and carbon footprints of llm-powered intelligent chatbots. *Engineering* **40**, 202–210 (2024).
246. Shrouf, F., Ordieres, J. & Miragliotta, G. Smart factories in Industry 4.0: a review of the concept and of energy management approached in production based on the Internet of Things paradigm. In *Proceedings of the 2014 IEEE International Conference on Industrial Engineering and Engineering Management*. 697–701 (IEEE, 2014).
247. Tibrewal, K. et al. Reconciliation of energy use disparities in brick production in India. *Nat. Sustain.* **6**, 1248–1257 (2023).

248. Cao, Y., Liu, S. & Deng, A. Digital carbon neutrality: a new way explored by industrial internet. *Environ. Dev. Sustain.* 1–18 (2024).
249. Qian, F., Tang, Y. & Yu, X. The future of process industry: a cyber–physical–social system perspective. *IEEE Trans. Cybern.* **54**, 3878–3889 (2023).
250. Kanoun, O. et al. Prospects of wireless energy-aware sensors for smart factories in the industry 4.0 era. *Electronics* **10**, 2929 (2021).
251. Li, M. et al. Artificial intelligence enabled self-powered wireless sensing for smart industry. *Chem. Eng. J.* **492**, 152417 (2024).
252. Perera, Y. S., Ratnaweera, D., Dasanayaka, C. H. & Abeykoon, C. The role of artificial intelligence-driven soft sensors in advanced sustainable process industries: a critical review. *Eng. Appl. Artif. Intell.* **121**, 105988 (2023).
253. Matsunaga, F., Zytkowski, V., Valle, P. & Deschamps, F. Optimization of energy efficiency in smart manufacturing through the application of cyber–physical systems and industry 4.0 technologies. *J. Energy Resour. Technol.* **144**, 102104 (2022).
254. Zhou, M., Qiao, Y., Liu, B., Vogel-Heuser, B. & Kim, H. Machine learning for industry 4.0. *IEEE Robot. Autom. Mag.* **30**, 8–9 (2023).
255. Zafar, M. H., Langås, E. F. & Sanfilippo, F. Exploring the synergies between collaborative robotics, digital twins, augmentation, and industry 5.0 for smart manufacturing: a state-of-the-art review. *Robot. Comput.-Integr. Manuf.* **89**, 102769 (2024).
256. Akundi, A. et al. State of Industry 5.0—analysis and identification of current research trends. *Appl. Syst. Innov.* **5**, 27 (2022).
257. Fraga-Lamas, P., Lopes, S. I. & Fernández-Caramés, T. M. Green IoT and edge AI as key technological enablers for a sustainable digital transition towards a smart circular economy: an industry 5.0 use case. *Sensors* **21**, 5745 (2021).
258. Tao, F., Zhang, H. & Zhang, C. Advancements and challenges of digital twins in industry. *Nat. Comput. Sci.* **4**, 169–177 (2024).
259. Liu, W. et al. The potential of smart factories in reducing environmental emissions: the evidence from Chinese listed manufacturing firms. *Humanit. Soc. Sci. Commun.* **11**, 1–16 (2024).
260. Zaiter, I. et al. Enabling industrial decarbonization: framework for hydrogen integration in the industrial energy systems. *Renew. Sustain. Energy Rev.* **203**, 114782 (2024).
261. Jamwal, A., Agrawal, R., Sharma, M. & Giallanza, A. Industry 4.0 technologies for manufacturing sustainability: a systematic review and future research directions. *Appl. Sci.* **11**, 5725 (2021).
262. Leng, J. et al. Review of manufacturing system design in the interplay of Industry 4.0 and Industry 5.0 (Part I): design thinking and modeling methods. *J. Manuf. Syst.* **76**, 158–187 (2024).
263. Huang, J., Wang, Y., Yang, A., Zhou, H. & Fan, C. Digitization and industrial low-carbon energy transition in China. *Energy* **322**, 135545 (2025).
264. Rojek, I., Mikołajewski, D., Mroziński, A. & Macko, M. Green energy management in manufacturing based on demand prediction by artificial intelligence—a review. *Electronics* **13**, 3338 (2024).
265. Abadi, M. M. K. F., Liu, C., Zhang, M., Hu, Y. & Xu, Y. Leveraging AI for energy-efficient manufacturing systems: review and future prospectives. *J. Manuf. Syst.* **78**, 153–177 (2025).
266. Hu, Y. & Man, Y. Energy consumption and carbon emissions forecasting for industrial processes: status, challenges and perspectives. *Renew. Sustain. Energy Rev.* **182**, 113405 (2023).
267. Tan, D., Suvarna, M., Tan, Y. S., Li, J. & Wang, X. A three-step machine learning framework for energy profiling, activity state prediction and production estimation in smart process manufacturing. *Appl. Energy* **291**, 116808 (2021).
268. Majeed, A. et al. A big data-driven framework for sustainable and smart additive manufacturing. *Robot. Comput.-Integr. Manuf.* **67**, 102026 (2021).
269. Liang, S., Yang, J. & Ding, T. Performance evaluation of AI driven low carbon manufacturing industry in China: an interactive network DEA approach. *Comput. Ind. Eng.* **170**, 108248 (2022).
270. Tseng, C.-J. & Lin, S.-Y. Role of artificial intelligence in carbon cost reduction of firms. *J. Clean. Prod.* **447**, 141413 (2024).
271. Mallapragada, D. S. et al. Decarbonization of the chemical industry through electrification: Barriers and opportunities. *Joule* **7**, 23–41 (2023).
272. Price, C. R., Nimbalkar, S. U., Thirumaran, K. & Cresko, J. Smart manufacturing pathways for industrial decarbonization and thermal process intensification. *Smart Sustain. Manuf. Syst.* **7**, 41–53 (2023).
273. Manikandan, S. et al. Artificial intelligence-driven sustainability: Enhancing carbon capture for. *Sustain. Dev. Goals* **33**, 2004–2029 (2025).
274. Xie, X., Fu, H., Zhu, Q. & Hu, S. Integrated optimization modelling framework for low-carbon and green regional transitions through resource-based industrial symbiosis. *Nat. Commun.* **15**, 3842 (2024).
275. Chinnathai, M. K. & Alkan, B. A digital life-cycle management framework for sustainable smart manufacturing in energy intensive industries. *J. Clean. Prod.* **419**, 138259 (2023).
276. Zhang, X., Khan, K., Shao, X., Oprean-Stan, C. & Zhang, Q. The rising role of artificial intelligence in renewable energy development in China. *Energy Econ.* **132**, 107489 (2024).
277. Liu, J., Lin, F., Li, X., Lim, K. H. & Zhao, S. Physics-informed LLM-agent for automated modulation design in power electronics systems. arXiv:2411.14214v1 (2024).
278. Dreyer, J. China made waves with Deepseek, but its real ambition is AI-driven industrial innovation. *Nature* **638**, 609–611 (2025).
279. Zhu, J. et al. Review and prospect of data-driven techniques for load forecasting in integrated energy systems. *Appl. Energy* **321**, 119269 (2022).
280. Machlev, R. et al. Explainable artificial intelligence (XAI) techniques for energy and power systems: review, challenges and opportunities. *Energy AI* **9**, 100169 (2022).
281. Shoaei, M., Noorollahi, Y., Hajinezhad, A. & Moosavian, S. F. A review of the applications of artificial intelligence in renewable energy systems: an approach-based study. *Energy Convers. Manag.* **306**, 118207 (2024).
282. Li, Y. et al. Artificial intelligence-based methods for renewable power system operation. *Nat. Rev. Electr. Eng.* **1**, 163–179 (2024).
283. Mullanu, S., Chua, C., Molnar, A. & Yavari, A. Artificial intelligence for hydrogen-enabled integrated energy systems: A systematic review. *Int. J. Hydrog. Energy* **141**, 283–303 (2025).
284. Zhou, Y. & Dan, Z. Modern energy resilience studies with artificial intelligence for energy transitions. *Cell Rep. Phys. Sci.* **6**,102508 (2025).
285. Bui, V.-H. et al. A critical review of safe reinforcement learning strategies in power and energy systems. *Eng. Appl. Artif. Intell.* **143**, 110091 (2025).
286. Han, H., Zhang, H., Yang, J. & Su, H. Distributed model predictive consensus control for stable operation of integrated energy system. *IEEE Trans. Smart Grid* **15**, 381–393 (2024).
287. Zhang, M. et al. Dual-stackelberg game-based trading in community integrated energy system considering uncertain demand response and carbon trading. *Sustain. Cities Soc.* **101**, 105088 (2024).
288. Li, S. et al. Physics-model-free heat-electricity energy management of multiple microgrids based on surrogate model-enabled multi-agent deep reinforcement learning. *Appl. Energy* **346**, 121359 (2023).
289. EUREF AG. *EUREF-Campus Berlin: Sustainable Urban Development* (Zukunftsorte Berlin, 2023).
290. Carvallo, C., Jalil-Vega, F. & Moreno, R. A multi-energy multi-microgrid system planning model for decarbonisation and decontamination of isolated systems. *Appl. Energy* **343**, 121143 (2023).
291. Lin, X., Sun, P., Zhong, W. & Wang, J. Thermodynamic analysis and operation investigation of a cross-border integrated energy system

based on steam Carnot battery. *Appl. Therm. Eng.* **220**, 119804 (2023).
292. Alabi, T. M., Lu, L. & Yang, Z. Real-time automatic control of multi-energy system for smart district community: a coupling ensemble prediction model and safe deep reinforcement learning. *Energy* **304**, 132209 (2024).
293. Van Dam, K. I. M. *Working towards a Positive Energy District for Business Park De Zwette, Leeuwarden: Exploring Stakeholder Perspectives* https://greenhubdenmark.dk/media/3bffmr4p/2025_03_06_final_report-exploring-stakeholder-perspectives_de-zwette-1.pdf (2025).
294. Kumar, S. et al. A multi-objective optimisation framework for a standalone hybrid offshore renewable energy system with electrical and hydrogen loads. *Energy* **330**, 136826 (2025).
295. Jhbdel. *File: Airfoil with Flow.Png*. Wikimedia https://commons.wikimedia.org/wiki/File:Airfoil_with_flow.png (2010).

Acknowledgements

This work is supported by the National Key R&D Program of China (Grant No. 2023YFE0108600) and the National Natural Science Foundation of China (Grant No. 51806190). This work is also supported by the Self-directed project, State Key Laboratory of Clean Energy Utilization.

Author contributions

X.L., Y.M., and W.Z. conceived the idea and developed the review framework. Y.M. and N.Z. wrote the manuscript. P.S. conducted data analysis and assisted in writing and editing the manuscript. All authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Xiaojie Lin.

Reprints and permissions information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026